

US ROBOTICS SPORTSTER OPERATING INSTRUCTIONS MANUAL



Quick Links

[External Modem Installation with Windows 3](#)

[Technical Quick Reference](#)

[Command Summary](#)

[Dip Switches \(Modems with Dip Switches Only\)](#)

Table of Contents

- Table of Contents
- Welcome to x2 Information Access
- Before You Begin (Windows 95 Users)
- Internal Modem Installation with Windows 3
- A Word about COM Ports and IRQs
- How to Use ComTest to Determine Your Modem's Settings
- Testing an Installed Modem
- What to Do with ComTest's Recommendation
- How to Change the Modem's Settings
- How to Insert the Modem into the Computer
- Internal Modem Installation with Windows 95
- Plug and Play Installation
- How to Prepare for Plug and Play Installation
- How to Insert the Modem into the Computer
- How to Move Through the "New Hardware Found" Screens
- External Modem Installation with Windows 3
- How to Connect the Modem to the Computer
- External Modem Installation with Windows 95
- How to Determine Which Serial Cable to Buy
- How to Connect the Modem to the Computer
- Software Installation and Testing
- Software Installation and Registration Using the Setup Wizard
- Sending Your First Fax
- Installing Other Fax/Data Software
- Type of Modem
- Initialization String
- Flow Control
- UART (External Modems Only)
- Using Modem Station
- Why Modem Station
- What Does Modem Station Do

Installing Modem Station

Starting Modem Station

Using Detect New Modems

Using Terminal

Using Modem Configurator

Using Modem to Computer

Using the Extended Information Screens

Robotics Modem Update Wizard

Installation

Operation

Troubleshooting and Online Help Resources

Online Help Resources

Are You Still Having Problems

If You Need to Return the Modem to Us

RapidComm Troubleshooting Tips

A Note to Users with Older Versions of RapidComm on Their Systems

Glossary

asynchronous transmission

binary digit

auto dial

bit rate

Technical Quick Reference

Front Panel Lights (External Modems)

Command Summary

DIP Switches (Modems with DIP Switches Only)

S-Registers

The Serial Interface

serial interface pin definitions

Regulatory Information

Manufacturer's Declaration of Conformity

Caution to the User

IC (Canada)

UL Listing/CUL Listing

Connecting to the Telephone Company

Fax Branding

[Radio and Television Interference](#)

[For Canadian Modem Users](#)

[Limited Warranty](#)

[Index](#)

[Other ManualsLib Projects](#)



This manual covers installation and operating instructions for the following U.S. Robotics modems:

Sportster® 56 kbps* internal and external modems

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Skokie, IL 60077-2690
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*** IMPORTANT!** All x2 products are capable of 56 kbps downloads; however, due to FCC rules which restrict power output of the service providers' modems, current download speeds are limited to 53 kbps. Actual speeds may vary depending on line conditions. Uploads from users to service providers travel at speeds up to 28.8 kbps. An x2-capable modem, an x2-compatible analog phone line and an x2-capable Internet Service Provider are necessary for these high-speed downloads.

Welcome to x2 Information Access	1
Product Features	3
Before You Begin (Windows 95 Users)	4
Internal Modem Installation with Windows 3.x	5
A Word about COM Ports and IRQs	5
How to Use ComTest to Determine Your Modem's Settings	7
Testing an Installed Modem	9
What to Do with ComTest's Recommendation	10
How to Change the Modem's Settings	12
How to Insert the Modem into the Computer	14
Internal Modem Installation with Windows 95	21
Plug and Play Installation	21
How to Prepare for Plug and Play Installation	21
How to Insert the Modem into the Computer	23
How to Move Through the "New Hardware Found" Screens	27
External Modem Installation with Windows 3.x	31
How to Determine Which Serial Cable to Buy	31
How to Connect the Modem to the Computer	32
External Modem Installation with Windows 95	35
How to Determine Which Serial Cable to Buy	35
How to Connect the Modem to the Computer	36
Software Installation and Testing	43
Software Installation and Registration Using the Setup Wizard	43
Sending Your First Fax	58

Installing Other Fax/Data Software	61
Type of Modem	61
Initialization String	61
Flow Control	61
UART (External Modems Only)	62
Using Modem Station	63
Why Modem Station?	63
What Does Modem Station Do?	63
Installing Modem Station	63
Starting Modem Station	66
Using Detect New Modems	69
Using Terminal	70
Using Modem Configurator	73
Using Modem to Computer	76
Using the Extended Information Screens	80
U.S. Robotics Modem Update Wizard	89
Installation	89
Operation	94
Troubleshooting and Online Help Resources	101
Online Help Resources	107
Are You Still Having Problems?	111
If You Need to Return the Modem to Us	111
RapidComm Troubleshooting Tips	112
A Note to Users with Older Versions of RapidComm on Their Systems	113

Glossary	114
Technical Quick Reference	127
Front Panel Lights (External Modems)	128
Command Summary	129
DIP Switches (Modems with DIP Switches Only)	144
S-Registers	146
The Serial Interface	157
Regulatory Information	159
Manufacturer's Declaration of Conformity	159
Caution to the User	160
IC (Canada)	160
UL Listing/CUL Listing	160
Connecting to the Telephone Company	161
Fax Branding	161
Radio and Television Interference	162
For Canadian Modem Users	163
Limited Warranty	166
Index	169

WELCOME TO X2™ INFORMATION ACCESS



The latest breakthrough in online communications

Until now, 33.6 kbps was thought to be the practical limit for speed over standard phone lines. Now, x2™ shatters that barrier, to bring you download speeds of up to 56 kbps. However, due to FCC rules which restrict

power output of your service provider's modems, current download speeds are limited to 53 kbps. This modem is software upgradable, allowing easy upgrades to new features and enhancements as they become available.

An integrated end-to-end solution from the leader

U.S. Robotics is well positioned to bring you x2 technology. Our Sportster® modems are the world's number one seller, and our Total Control™ and Courier™ product lines are favored by many Internet service providers. This widespread acceptance allows U.S. Robotics to offer x2 technology to virtually anyone.

WELCOME TO X2 INFORMATION ACCESS

Taking advantage of the modern telephone network

x2 technology takes advantage of the typical network configuration found when an analog subscriber connects to a digitally connected server. Because it requires no analog-to-digital conversions in the downstream path (which can cause line noise), x2 can use nearly all of the available 64K network bandwidth. (Upstream data, typically less speed sensitive, travels at the standard V.34 rate.)

The result is a completely new kind of transmission technique. Based on “encoding” rather than “modulation,” it can give you download speeds that you never thought possible. What’s more, with standard V.42 *bis*

compression, x2 can download at speeds up to a blistering 115.2 kbps.

Updating x2 technology is easy and FREE!

Read the section of this manual titled “U.S. Robotics Modem Update Wizard” (page 85) for information about using this software (on the *Connections*[™] CD) to update your modem’s code to the latest version.

The new standard for online speed

U.S. Robotics has already submitted x2 to the ITU-T standards committee for acceptance as the next online standard. For more information on U.S. Robotics’ x2 technology, see our World Wide Web page at <http://x2.usr.com>.

PRODUCT FEATURES

Your new x2 modem provides many advanced features. Here are just a few:

Modulation Schemes

x2 technology
ITU-T V.34+
ITU-T V.34
ITU-T V.32bis
ITU-T V.32
ITU-T V.22bis
ITU-T V.22
ITU-T V.23
Bell 212A
ITU-T V.21
Bell 103

Error Control and Data Compression Schemes

ITU-T V.42
ITU-T V.42bis
MNP 2-5

Fax Modulation Schemes

ITU-T V.17
ITU-T V.29
ITU-T V.27ter
ITU-T V.21

Fax Standards

EIA 578 Class 1 FAX
EIA 592 Class 2.0 FAX

Front Channel Link Rates

33333, 37333, 41333, 42666,
44000, 45333, 46666, 48000,
49333, 50666, 52000, 53333,
54666, 56000, 57333

Back Channel Link Rates

14400, 16800, 19200, 21600,
24000, 26400, 28800, 31200

V.34+ Link Rates

4800, 7200, 9600, 12000, 14400,
16800, 19200, 21600, 24000,
26400, 28800, 31200, 33600

V.32bis Link Rates

4800, 7200, 9600, 12000, 14400

Additional Link Rates

300, 1200/75 (V.23), 1200, 2400

Fax Link Rates

2400, 4800, 7200, 9600, 12000,
14400

BEFORE YOU BEGIN (WINDOWS 95 USERS)

Your Sportster modem is a Plug and Play device. Windows 95 can automatically identify a Plug and Play device and determine if your system has the resources necessary to support the device. However, Plug and Play will not work if you do not have resources available or if devices on your system are not reporting resource usage correctly. Here's how you can verify that your system has the necessary resources before installing the modem.

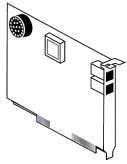
1. Click the Windows 95 **Start** button, point to **Settings**, and then click **Control Panel**.
2. Double-click the **System** icon.
3. When the "System Properties" screen appears, click the **Device Manager** tab.
4. Double-click **Computer** and the "Computer Properties" screen appears.

Select the option at the top of the screen to show Interrupt Requests (IRQs).

You will see the IRQs your system is currently using. If IRQs 3, 4, 5, and 7 are being used, you need to free an IRQ before you begin installation. This process involves moving a device from the IRQ you want to use to a different (and unusually higher) IRQ setting. Please read the documentation for (or contact the manufacturer of) the device that is currently using the IRQ you want to use for your modem to learn more about what you should do to free the IRQ for your modem.

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

You'll need these items from your Sportster modem box:



modem



phone cord



Connections CD

Plus:



a screwdriver (not included)

A Word about COM Ports and IRQs

Most computer accessories — a mouse, sound card, enhanced graphics card, scanner, etc. — require a special connection through which they can communicate with the computer. For some devices, this connection is called a communications (or COM) port. Most computers have 1 or 2 COM ports, but they can have up to 4. Although internal modems do not connect to COM ports directly, they do require a COM port setting, which is determined by the setting on the modem's COM jumper pins.

Each COM port uses an interrupt request (IRQ). An accessory uses an IRQ to get the attention of your computer's central processing unit (CPU) so that the accessory can perform

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

a task. The computer stops what it's doing, depending on the priority of the request, to help the accessory perform its task. When two accessories share an IRQ, it's like two people asking different questions in unison to a third person. Just as the person being asked the two questions cannot understand either request, a computer can lock up or otherwise fail to communicate properly with your modem when there is an IRQ conflict.

D KEY POINT: Accessories cannot share COM ports and should not share IRQs. When accessories try to share settings, they will either not work properly or not work at all. This section of the manual will help you understand the COM/IRQ settings on your new Sportster modem.

U.S. Robotics has set your modem to a default Plug and Play setting geared towards Windows 95 users. In this configuration, the shunts used to set your modem's COM port and IRQ settings are hanging from single jumper pins and will not affect your modem's settings. As a Windows 3.x user, you need to run the ComTest program (on the *Connections* CD) to determine what settings your modem should use.



NOTE: Some communications software programs require a particular setting for your modem (RapidComm, which shipped with your modem, does not). If you wish to use a program other than RapidComm, now is a good time to read that software's manual to determine what setting is required.

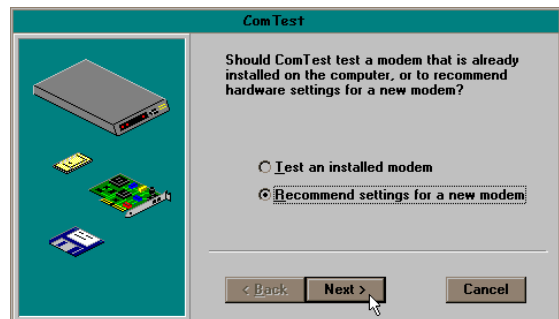
INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

How to Use ComTest to Determine Your Modem's Settings

1. Insert the *Connections* CD into your CD-ROM drive.
 2. In Windows' Program Manager, click **Run** on the **File** menu. Type **d:\comtest.exe** and press **ENTER**. This starts ComTest, the program that determines which COM ports and IRQs are available for use by your modem.
 3. The screen in the next column appears when ComTest starts.
- If there is a modem in your computer which you are replacing with your new Sportster modem, go to "Testing an Installed Modem" on page 9 to determine which

COM and IRQ settings the older modem is using.

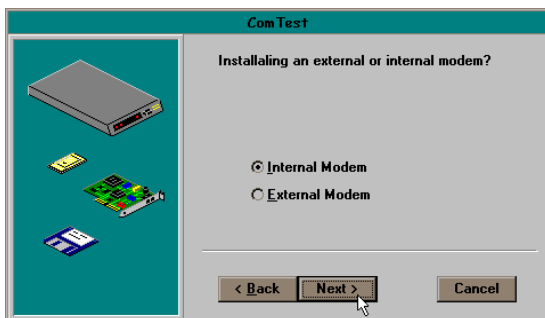
- If a modem is not currently installed in your computer, click **Recommend settings for a new modem**.



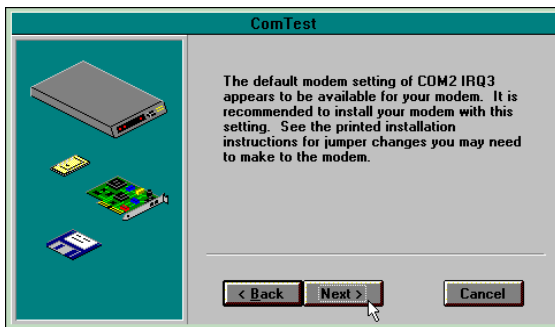
Then click **Next**.

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

4. Click **Internal Modem**. Then click **Next**.



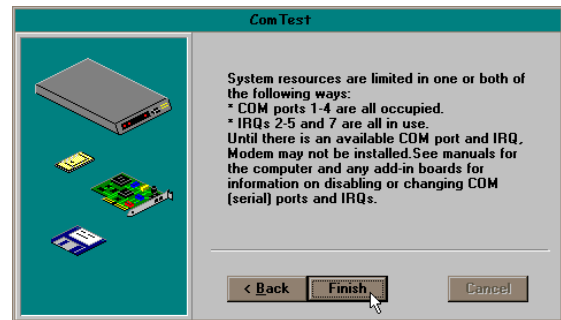
5. If you have a free setting, you will see a screen like this.



COM2/IRQ3 is free in this example. ComTest might instead recommend COM1/IRQ4, COM3/IRQ4, or COM4/IRQ3 for your modem.

Write down the displayed settings in the spaces provided on the first page of this manual. You'll need to know these settings later. Click **Next**.

If you do not have a free setting, you will see a screen like this.



INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

Click **Finish** to exit ComTest. Go to “What to Do with ComTest’s Recommendation” on page 10.

6. When you see this screen, click **Finish** to exit ComTest.



Go to “What to Do with ComTest’s Recommendation” on page 10.

Testing an Installed Modem

If there is a modem already installed in your computer, you can determine its COM and IRQ settings by selecting the *Test an installed modem* option. Follow the instructions on screen until you see the screen that tells you “Testing is complete.” This screen will also tell you which COM and IRQ settings your present modem is using. These are the settings you want to use for your new Sportster modem. Write the settings down in the spaces provided on the first page of this manual.

Turn off and unplug your computer and remove your present modem. Go to “How to Change the Modem’s Settings” on page 12.

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

What to Do with ComTest's Recommendation



NOTE: If your communications software requires a modem that ComTest does not recommend, there is a good chance that the setting is being used by another device in your system. To free that setting in your system, consult your computer's manual. Go to "How to Change the Modem's Settings" on page 12.

If ComTest recommended COM1/IRQ4

Write "COM1" and "IRQ4" in the spaces provided on the first page of this manual. Go to "How to Change the Modem's Settings" on page 12.

If ComTest recommended COM2/IRQ3

Write "COM2" and "IRQ3" in the spaces provided on the first page of this manual. Go to "How to Change the Modem's Settings" on page 12.

If ComTest recommended COM3/IRQ4 or COM4/IRQ3

Do not use either of these settings. While the COM port part of the setting is acceptable (ComTest only recommends COM ports that are not being used), the IRQ part of the setting is not acceptable. When ComTest recommends either COM3/IRQ4 or COM4/IRQ3, the IRQ in the setting is being used by another accessory. You could use the suggested IRQ for the modem, but you run the risk that the modem and/or the other accessory sharing the IRQ might not work properly.

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

If you do not have a sound card, use COM3/IRQ5. Write “COM3” and “IRQ5” in the spaces provided on the first page of this manual. You will have to change the settings on your modem. Go to “How to Change the Modem’s Settings” on page 12.

If you do have a sound card, use COM2/IRQ3. To use this setting, you have to first disable your computer’s second serial port (COM2). This is a pronged socket on the back of your computer.

- Go to your computer manufacturer’s manual.
- Find out which of the sockets on the back of your computer is the second serial port.
- If you have something plugged into that port, find out if you can plug it in somewhere else. Any accessory plugged

into that port will not work after the port has been disabled.

- Find out how to disable the COM port.
- Then return to this point in this manual to continue.

After disabling COM2, you can use the COM2/IRQ3 setting.

Write “COM2” and “IRQ3” on the first page of this manual and go to “How to Change the Modem’s Settings” on page 12.

If ComTest reports that You do not have any available COM ports and/or IRQs

We recommend you disable COM1 or COM2. When you disable one of these COM ports, you can use it and its default IRQ for your modem. Go to your computer manufacturer’s manual.

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

- Find out which socket is COM1 and which is COM2.
- If nothing is plugged into either port, you can choose either of the ports to disable. Your computer manufacturer's manual will tell you how to disable the COM port.
- If one port does not have anything plugged into it, note if the port is COM1 or COM2. This is the port you'll want to disable for your modem.
- If both ports are being used, you may be able to attach one of the plugged-in accessories elsewhere so that you can disable its COM port. The accessory will not work once its port is disabled.

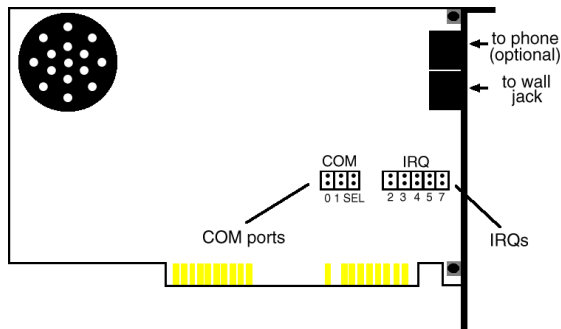
If you've disabled COM1, you can now use the COM1/IRQ4 setting. Write it down on the first page of this manual. Go to "How to Change the Modem's Settings" (on this page).

If you've disabled COM2, you can now use the COM2/IRQ3 setting. Write the setting down on the first page of this manual.

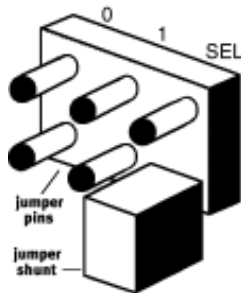
How to Change the Modem's Settings

1. Always touch an unpainted metal part of your computer (the back is usually unpainted) to discharge static electricity before handling the modem. Static can damage your modem. Then take the modem out of its plastic bag.
2. Find the COM and IRQ jumper shunts (small black plastic pieces) on your modem's jumper pins (see the following diagram).

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X



The jumper shunts will be hanging from single jumper pins as shown in the following illustration.



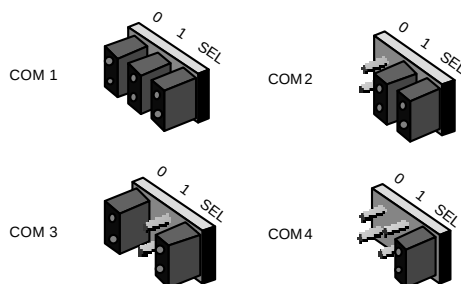
3. To change the COM port and IRQ settings, you need to reposition the jumper shunts on the COM port and IRQ pins. To do this, lift the jumper shunts off the pins.



TIP: Grasp the jumper shunts with a tweezers or needle-nosed pliers. **DO NOT** grasp too firmly, as you may crush the shunts. If a jumper shunt seems stuck, gently rock it back and forth as you lift. Do not touch any other part of the modem or your computer with the tweezers/pliers. A shunt needs to be sitting on both jumper pins in order to effectively set the modem to the desired setting.

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

4. Move the jumper shunt to the new setting.
- The COM port setting can involve one to three shunts. The four possible COM port settings are as follows:



- Unlike most COM port settings, the IRQ setting involves only one jumper shunt. Simply move the jumper shunt to the pins labeled with the IRQ you need.

How to Insert the Modem into the Computer



NOTE: Before installing your modem, write its serial number in the space provided on the first page of this manual. (You'll find the serial number underneath the bar code on the white sticker on the modem and on the outside of the box the modem came in.) If you ever need to call our customer support department, a customer support representative will ask you for the serial number. This will help him or her identify your modem.

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

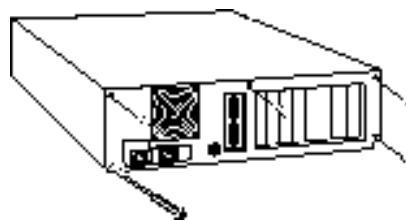
1. Turn off your computer and unplug it from the electrical outlet.
2. Unplug any peripheral devices (printer, monitor, keyboard, mouse, etc.) from the computer.
3. Remove the screws from your computer's cover and then remove the cover, as shown in the following diagrams. Your computer may differ in appearance from these diagrams, but the basic principle for removing the cover should be the same. Contact your computer manufacturer or review their manual if you need further instructions.



TIP: Before you unplug anything, label the cords or make a sketch of how things are connected. This can be helpful when you plug things back in later.



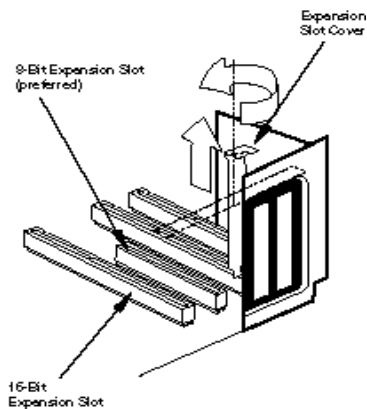
CAUTION: To avoid the risk of electric shock, make sure your computer and all peripheral devices are turned off and unplugged.



INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X



4. Find an empty ISA expansion slot at least as long as the gold edge of your modem. (ISA slots have black plastic grooves lined with silver.) Unscrew and remove the expansion slot cover (the long narrow piece of metal that keeps dust from entering through the opening perpendicular to the slot). Be careful not to drop the screw into the computer. You will need it later to screw the modem into place.

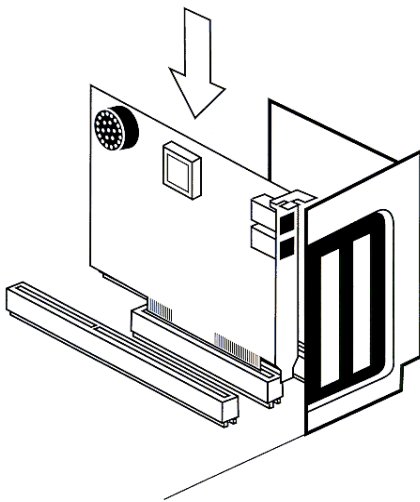


5. Holding the modem at each corner, with the gold edge facing the slot, push the modem down as gently as possible until it snaps into the expansion slot. (NOTE: The drawings show horizontally aligned expansion slots. Some computers have vertically aligned slots. The instructions apply to both styles.)

You need to apply a little pressure to seat the modem properly. Sometimes a gentle back-and-forth motion helps to fit the

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

modem all the way into the slot. If you feel resistance, the modem may not be properly lined up with the slot. Do not force it into the slot. Remove the modem and try again.



6. Once the modem is in place, fasten it firmly into place using the screw that you removed in step 4.

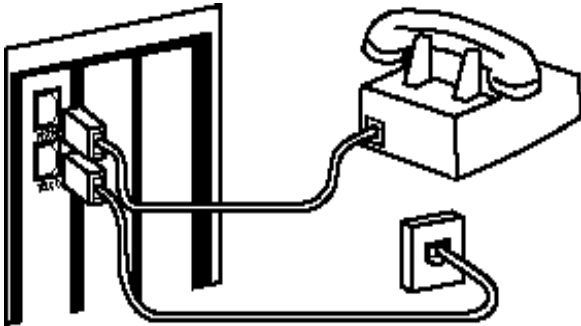
7. Replace the computer's cover and fasten it with the screws you removed in step 3.
8. If you currently have a phone plugged into the wall jack you plan to use for the modem, disconnect the phone's cord from the jack.

WARNING : The phone jack you use must be for an **ANALOG** phone line (the type found in most homes). Many office buildings have digital phone lines. Be sure you know which type of line you have. The modem will be damaged if you use a digital phone line.

9. Plug one end of the phone cord that came with the modem into the TELCO jack at the rear of the modem. Plug the other end of the cord into the wall jack.

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

- 10.** If you wish to use a phone through the same phone wall jack as the modem when the modem is not in use, plug your phone's cord into the modem's PHONE jack.

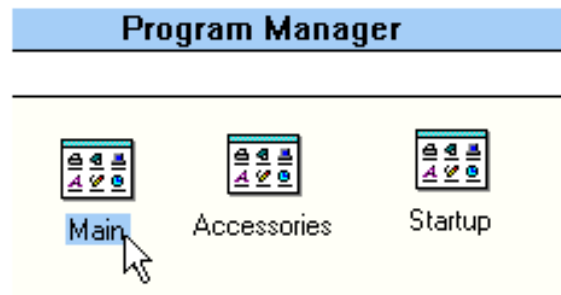


NOTE: You cannot use the modem and a phone at the same time if they share the same telephone line.

- 11.** Plug the power cords, cables, and peripherals back into the computer and turn the computer on.

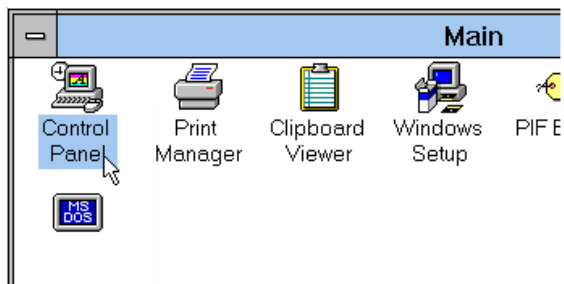
- 12.** Start Windows.

- 13.** If you have your modem set to COM1/IRQ4 or COM2/IRQ3, go to "Software Installation and Testing" on page 43. If you have your modem configured to any other setting, open Program Manager and double-click the **Main** icon.

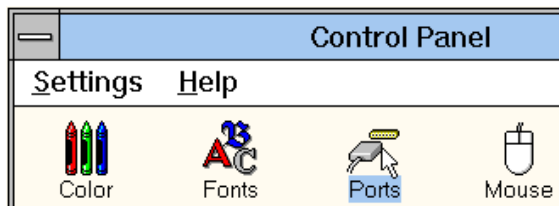
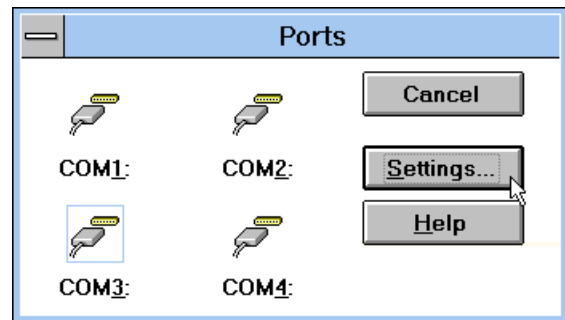


- 14.** Double-click the **Control Panel** icon.

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X



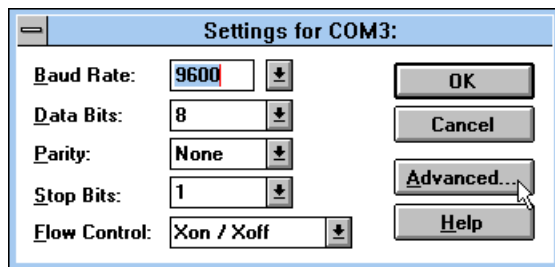
15. Double-click the **Ports** icon.



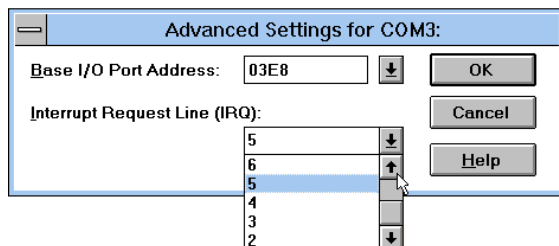
16. If it's not already selected, click the COM port for your modem. Then click **Settings**.

INTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

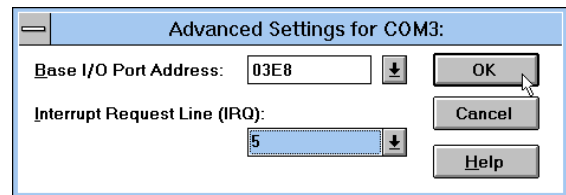
17. Click **Advanced**.



18. In the **Interrupt Request Line (IRQ)** box, select the IRQ that ComTest recommended.



19. Click **OK**.

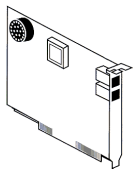


You will see a brief series of screens after the preceding screen. Click **OK** on each screen until you see a screen with a **Close** button. Click **Close** to exit Program Manager.

You are now ready to install the software on the *Connections* CD-ROM (included with your Sportster modem). Turn to "Software Installation and Testing" (page 43) for detailed information about installing the software, registering the modem, and trying out your modem by faxing U.S. Robotics.

INTERNAL MODEM INSTALLATION WITH WINDOWS 95

You'll need these items from your Sportster modem box:



modem



phone cord



Connections CD

Plus:



a screwdriver (not included)

Plug and Play Installation

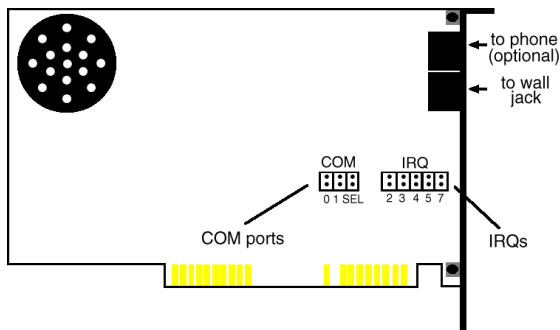
With Windows 95, you may be able to simply plug your new modem into your computer, power up, and be ready to work. The operating system should do all the work of recognizing the modem for you. This section will guide you through this “Plug and Play” installation.

How to Prepare for Plug and Play Installation

1. Touch an unpainted metal part of your computer (the back is usually unpainted) to discharge static electricity. Static can damage your modem.
2. Take the modem out of its plastic bag.

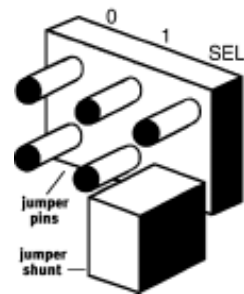
INTERNAL MODEM INSTALLATION WITH WINDOWS 95

- Find the jumper shunts (small black plastic pieces) on the COM port and IRQ jumper pins on your modem. They should be in roughly the area indicated in the following illustration.



When doing a “Plug and Play” installation, the jumper shunts should be placed so that they are hanging from single jumper pins rather than on pairs of pins. Your modem’s jumper shunts should already be in this Plug and Play configuration. If they are not,

move them so they are hanging as in the following diagram.



NOTE: If you need to move the shunts, grasp them with a tweezers or a needle-nosed pliers. **DO NOT** grasp too firmly, however, or you may crush the shunts. If a shunt seems stuck, try gently rocking it back and forth as you lift.

INTERNAL MODEM INSTALLATION WITH WINDOWS 95

How to Insert the Modem into the Computer



NOTE: Before installing your modem, write its serial number in the space provided on the first page of this manual. (You'll find the serial number underneath the bar code on the white sticker on the modem and on the outside of the box the modem came in.) If you ever need to call our customer support department, a customer support representative will ask you for the serial number. This will help him or her identify your Sportster modem.

1. Turn off your computer and unplug it from the electrical outlet.
2. Unplug any peripheral devices (printer, monitor, keyboard, mouse, etc.) from the computer.



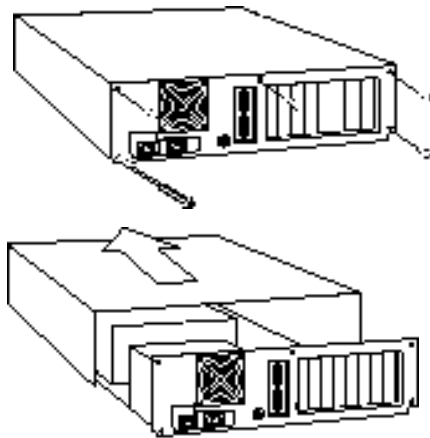
TIP: Before you unplug anything, label the cords or make a sketch of how things are connected. This can be helpful when you plug things back in later.



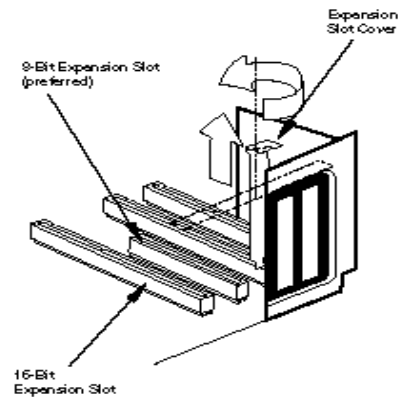
CAUTION : To avoid risk of electric shock, make sure your computer and all peripheral devices are turned off and unplugged from electrical outlets.

INTERNAL MODEM INSTALLATION WITH WINDOWS 95

3. Remove the screws from your computer's cover and then remove the cover, as shown in the following diagrams. Your computer may differ in appearance from these diagrams, but the basic principle for removing the cover should be the same. Contact your computer manufacturer or review their manual if you need further instructions.



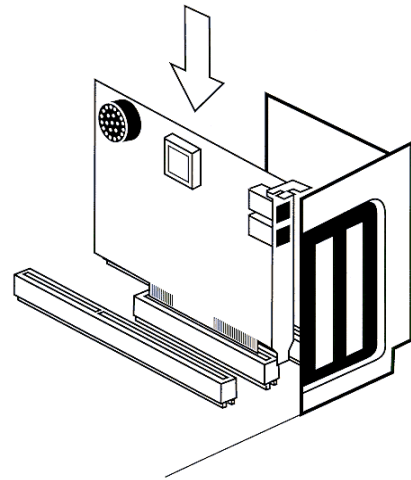
4. Find an empty ISA expansion slot that's at least as long as the gold edge of your modem. (ISA slots are black plastic grooves lined with silver.) Unscrew and remove the expansion slot cover (the long narrow piece of metal that keeps dust from entering through the opening perpendicular to the slot). Be careful not to drop the screw into the computer. You will need it later to screw the modem into place.



INTERNAL MODEM INSTALLATION WITH WINDOWS 95

5. Holding the modem at each corner, with the gold edge facing the slot, push the modem down as gently as possible until it snaps into the expansion slot. (Note: The following diagram shows horizontally aligned expansion slots. Some computers have vertically aligned slots. The instructions apply to both styles.)

You need to apply a little pressure to seat the modem properly. Sometimes a gentle back-and-forth motion helps fit the modem all the way into the slot. If you feel resistance, the modem may not be properly lined up with the slot. Do not force it into the slot. Remove the modem and try again.



6. Once the modem is in the slot, screw the modem firmly into place using the screw that you removed in step 4.
7. Replace the computer's cover and fasten it with the screws you removed in step 3.
8. Locate the wall jack you plan to use for the modem. If you have a phone plugged into

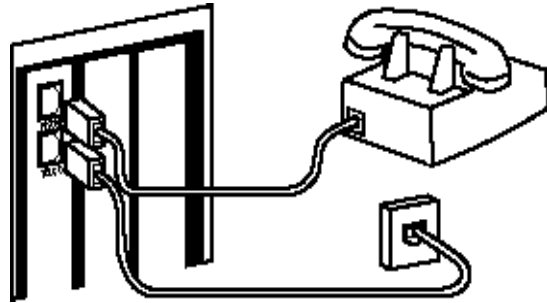
INTERNAL MODEM INSTALLATION WITH WINDOWS 95

this jack, disconnect the phone's cord from the jack.



WARNING : The phone jack you use must be for an ANALOG phone line (the type found in most homes). Many office buildings have digital phone lines. Be sure you know which type of line you have. The modem will be damaged if you use a digital phone line.

9. Plug one end of the phone cord included with the modem into the TELCO jack at the rear of the modem. Plug the other end of the cable into the wall jack.
10. If you wish to use a phone through the same phone wall jack as the modem when the modem is not in use, plug your phone's cord into the modem's PHONE jack.



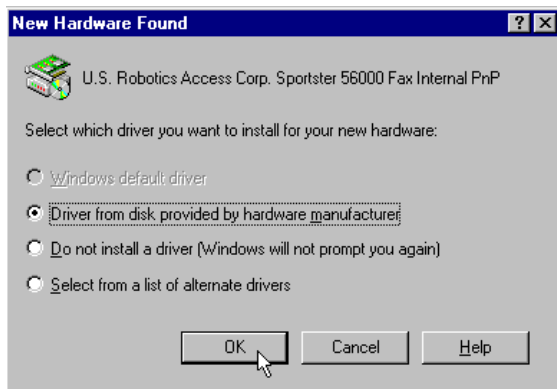
NOTE: You cannot use the modem and a phone at the same time if they share the same telephone line.

11. Plug the power cords, cables, and peripherals back into the computer and turn the computer on.

INTERNAL MODEM INSTALLATION WITH WINDOWS 95

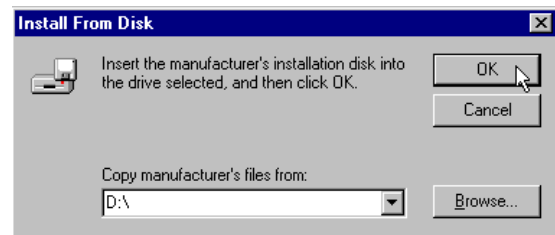
How to Move Through the “New Hardware Found” Screens

1. When Windows 95 restarts, the following screen appears automatically after a few seconds.



Click **Driver from disk provided by hardware manufacturer**. Then click **OK**.

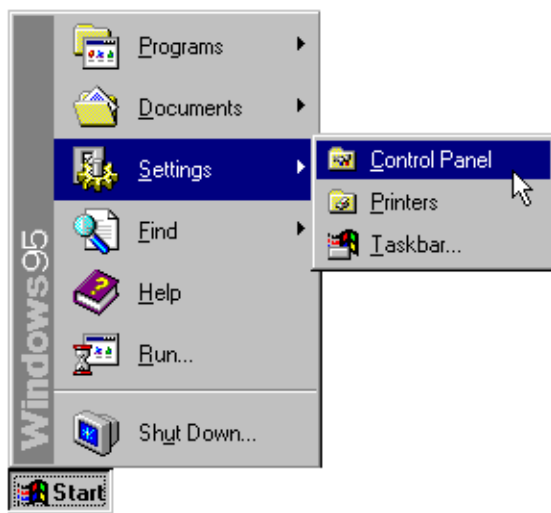
2. When you see this screen, insert the *Connections* CD into your CD-ROM drive and type **D:** to replace the **A:**. (NOTE: If your CD-ROM drive has a different letter name, type that letter instead of **D**.)



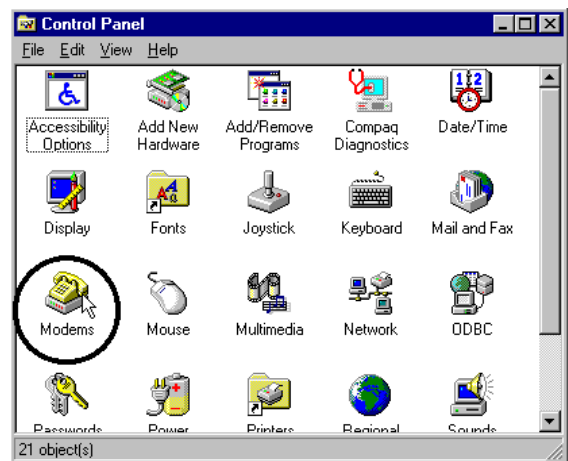
Click **OK**. Windows will load the modem's setup information.

INTERNAL MODEM INSTALLATION WITH WINDOWS 95

3. Once Windows 95 finishes loading the information from the CD-ROM, you should verify that the modem installation was a success. When your desktop returns, click the Windows' **Start** button and then point to **Settings**. Then click **Control Panel**.

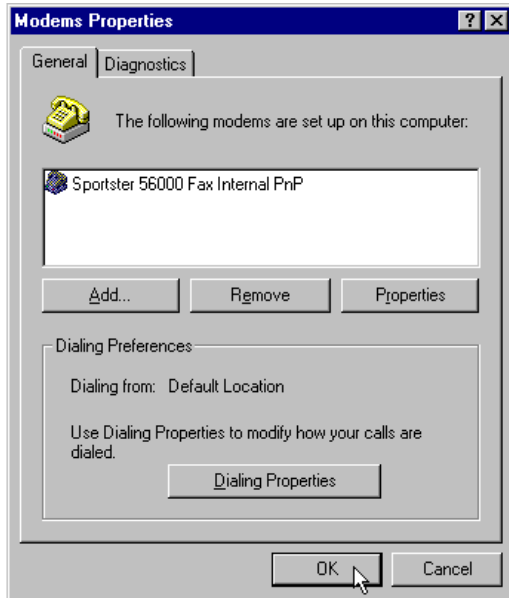


4. Double-click the **Modems** icon (circled in the screen image below).



INTERNAL MODEM INSTALLATION WITH WINDOWS 95

5. In the “Modems Properties” screen, you should see “Sportster 56000 Fax Internal PnP” listed.



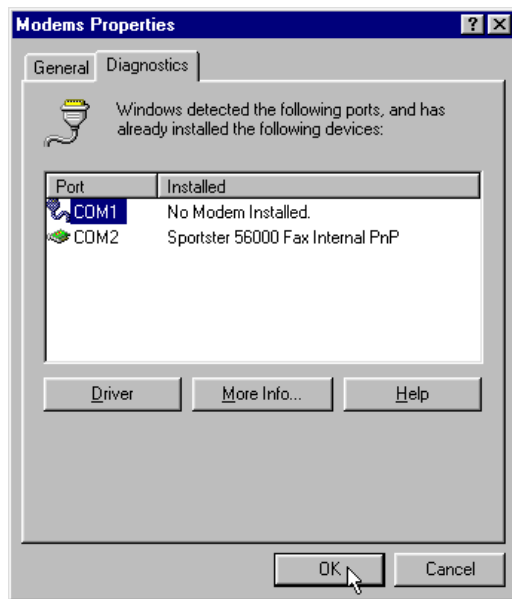
This means the installation was a success. Click **OK**.



NOTE: If you do not see your modem listed in the preceding screen, the installation was unsuccessful. Please refer to the “Troubleshooting and Online Help Resources” section (page 81) for possible solutions to your installation problem.

6. Next, click the **Diagnostics** tab at the top of the “Modems Properties” screen. Write down the COM setting for your modem. (Note: Your screen may show a different setting than that shown in the following screen.) You will need to know this setting when you install RapidComm later.

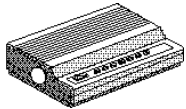
INTERNAL MODEM INSTALLATION WITH WINDOWS 95



Turn to “Software Installation and Testing” (page 43) for detailed information about installing RapidComm, registering the modem, and trying out your modem by faxing U.S. Robotics.

EXTERNAL MODEM INSTALLATION WITH WINDOWS 3.x

You'll need these items from your Sportster modem box:



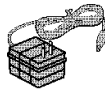
modem



phone cord

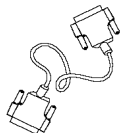


Connections CD



power adapter

Plus:



Serial modem cable. (Not included. See next column.)

How to Determine Which Serial Cable to Buy

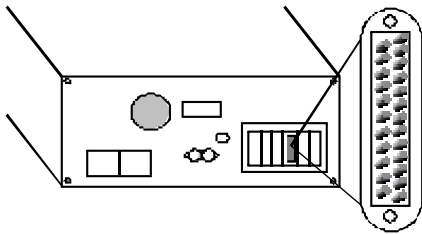
Before you can begin installation, you need to purchase an RS-232 serial modem cable. This section will help you to determine which serial modem cable to purchase.



NOTE: Your computer's rear panel may not be identical to the following diagrams. However, you should look for connectors that match the enlarged portions of the diagrams. If you cannot locate the proper connector, refer to your computer's manual.

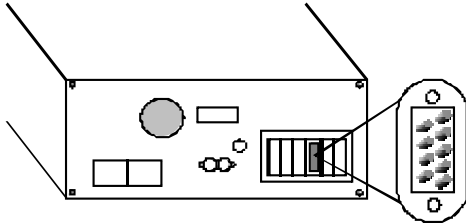
EXTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

If your PC's serial connector looks like this...



you need a shielded serial modem cable with a 25-pin female to 25-pin male connector.

If your PC's serial connector looks like this...



you need a shielded serial modem cable with a 9-pin female to 25-pin male connector.

How to Connect the Modem to the Computer

1. Turn off your computer and any attached devices, such as a printer.
2. Connect the serial cable to the modem and to the computer. When looking for the serial port label on the back of your computer, look for ports labeled COM, MODEM, RS-232, or SERIAL. **DO NOT** select AUX, GAME, LPT, or PARALLEL.
3. Plug the power adapter into the a standard wall outlet and insert its plug into the power jack on the modem (labeled with a power adapter icon on the bottom of the case).
4. Plug one end of the phone cord into the TELCO jack (labeled with a wall plug icon

EXTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

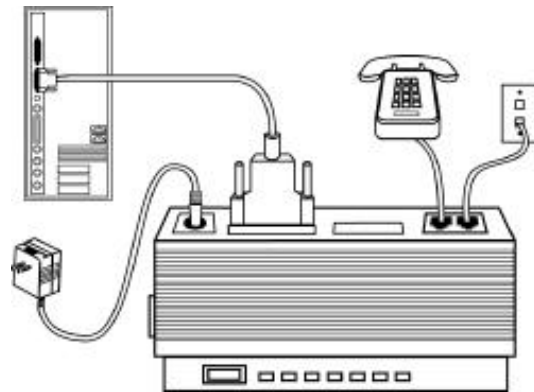
on the bottom of the case) and the other end into a phone wall jack.

 **WARNING :** The phone jack you use must be for an **ANALOG** phone line (the type found in most homes). Many office buildings have digital phone lines. Be sure you know which type of line you have. The modem will be damaged if you use a digital phone line.

5. If you wish to use your modem and phone through the same phone wall jack, plug your phone's cord into the modem's phone jack (labeled with a phone icon on the bottom of the case).



NOTE: You cannot use the modem and a phone at the same time if they share the same telephone line.



6. Turn your modem on. Then turn your computer on.

EXTERNAL MODEM INSTALLATION WITH WINDOWS 3.X

You are now ready to install the software on the *Connections* CD-ROM (included with your Sportster modem). Turn to “Software Installation and Testing” (page 43) for detailed information about installing

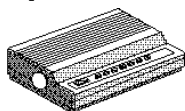
the software, registering the modem, and trying out your modem by faxing U.S. Robotics.

EXTERNAL MODEM INSTALLATION WITH WINDOWS

95

35

You'll need these items from your Sportster modem box:



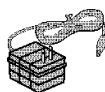
modem



phone cord

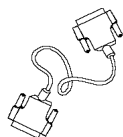


Connections CD



power adapter

Plus:



Serial modem cable. (Not included. See next column.)

How to Determine Which Serial Cable to Buy

Before you can begin installation, you need to purchase an RS-232 serial modem cable. This section will help you to determine which serial modem cable to purchase.

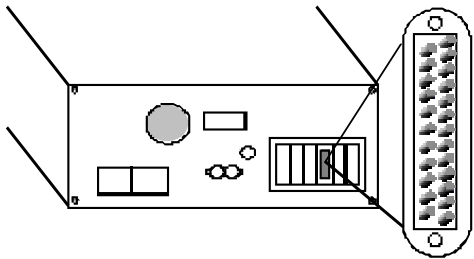


NOTE: Your computer's rear panel may not be identical to the following diagrams. However, you should look for connectors that match the enlarged portions of the diagrams. If you cannot locate the proper connector, refer to your computer's manual.

35

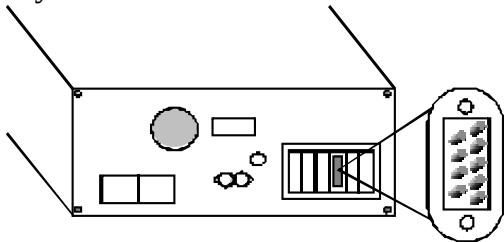
EXTERNAL MODEM INSTALLATION WITH WINDOWS 95

If your PC's serial connector looks like this...



you need a shielded serial modem cable with a 25-pin female to 25-pin male connector.

If your PC's serial connector looks like this...



you need a shielded serial modem cable with a 9-pin female to 25-pin male connector.

How to Connect the Modem to the Computer

1. Turn off your computer and any attached devices, such as a printer.
2. Connect the serial cable to the modem and to the computer. When looking for the serial port on the back of your computer, look for labels marked COM, MODEM, RS-232, or SERIAL. **DO NOT** select AUX, GAME, LPT, or PARALLEL.
3. Plug the power adapter into a standard wall outlet and insert its plug into the power jack on the modem (labeled with a power adapter icon on the bottom of the case).
4. Plug one end of the phone cord into the TELCO jack (labeled with a wall plug icon on the bottom of the case) and the other end into a phone wall jack.

EXTERNAL MODEM INSTALLATION WITH WINDOWS 95

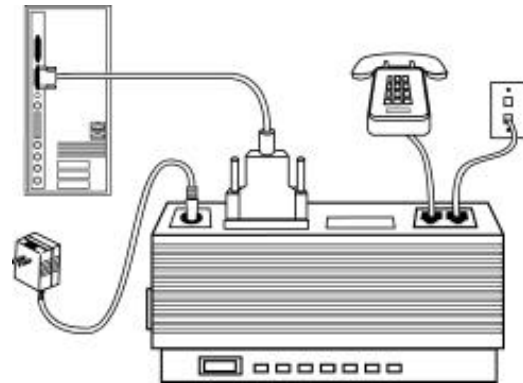


WARNING : The phone jack you use must be for an **ANALOG** phone line (the type found in most homes). Many office buildings have digital phone lines. Be sure you know which type of line you have. The modem will be damaged if you use a digital phone line.

5. If you wish to use your modem and phone through the same phone wall jack, plug your phone's cord into the modem's phone jack (labeled with a phone icon on the bottom of the case)



NOTE: You can not use the modem and phone at the same time if they share the same telephone line.



6. Turn on your modem (the power switch is located on the front panel). Then turn on your computer and peripherals.

How to Move Through the "New Hardware Found " Screens

EXTERNAL MODEM INSTALLATION WITH WINDOWS 95

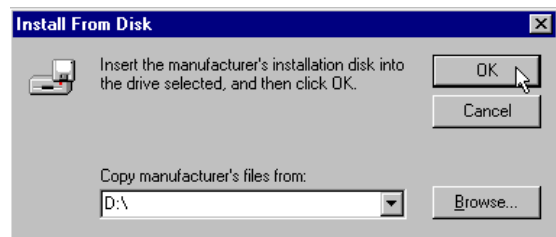
1. When Windows restarts, the following screen appears automatically after a few seconds.



Click **Driver from disk provided by hardware manufacturer**. Then click **OK**.

2. When you see this screen, insert the *Connections* CD into your CD-ROM drive and type **D:** to replace the **A:**. (NOTE: If your CD-ROM drive has a

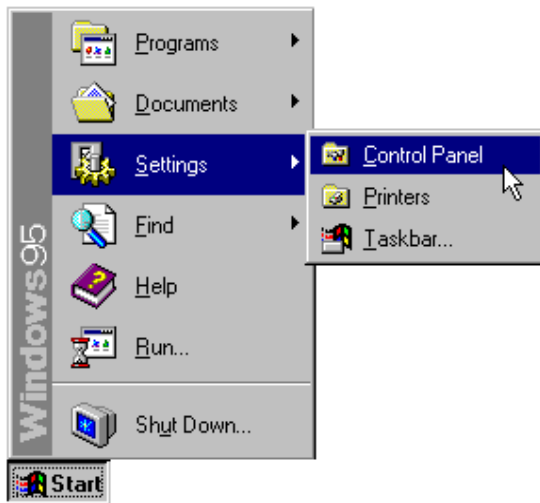
different letter name, type that letter instead of **D**.)



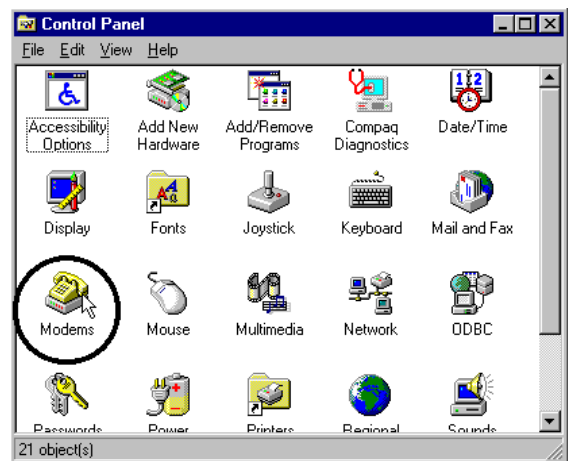
Click **OK**. Windows will load the modem's setup information.

EXTERNAL MODEM INSTALLATION WITH WINDOWS 95

3. Once Windows finishes loading the information from the CD-ROM, you should verify that the modem installation was a success. When your desktop returns, click the Windows **Start** button and point to **Settings**. Then click **Control Panel**.



4. Double-click the **Modems** icon (circled in the screen image below).



EXTERNAL MODEM INSTALLATION WITH WINDOWS 95

5. In the “Modems Properties” screen, you should see “Sportster 56000 Fax External PnP” listed.



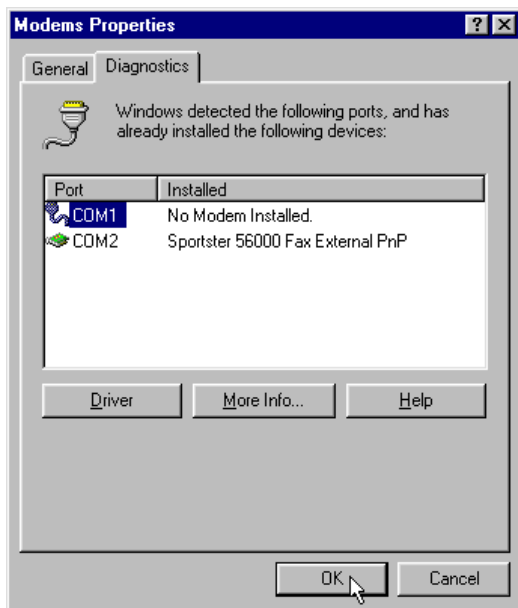
This means the installation was a success. Click **OK**.



NOTE: If you do not see your modem listed in the preceding screen, the installation was unsuccessful. Please refer to the “Troubleshooting and Online Help Resources” section (page 81) for possible solutions to your installation problem.

6. Next, click the **Diagnostics** tab at the top of the “Modems Properties” screen. Write down the COM setting for your modem. (Note: Your screen may show a different setting than that shown in the following screen.) You will need to know this setting when you install RapidComm later.

EXTERNAL MODEM INSTALLATION WITH WINDOWS 95



Turn to “Software Installation and Testing” (page 43) for detailed information about installing the software, registering the modem, and trying out your modem by faxing U.S. Robotics.

SOFTWARE INSTALLATION AND TESTING

Software Installation and Registration Using the Setup Wizard

This section guides you through the U.S. Robotics Setup Wizard, the *Connections*[™] CD-ROM interface, and the installation of RapidComm[™] (fax/data software). It also shows you how to register and test your new Sportster modem.

RapidComm fax/data software simplifies sending and receiving faxes directly from your computer desktop. You can build your own fax directory, send faxes to specified groups of phone numbers, request individual cover pages when necessary, and send individual faxes without exiting your word processing program.

Additionally, RapidComm lets you connect to BBSs and other online data providers. Take advantage of this access to enter a new world of information and entertainment.



NOTE: The following instructions apply to Windows 3.x and Windows 95 users. However, only Windows 95 screens are shown.



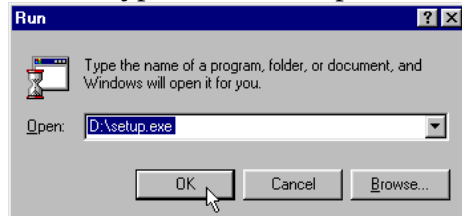
NOTE: If you have an older version of RapidComm installed on your system, uninstall it before continuing.

SOFTWARE INSTALLATION AND TESTING

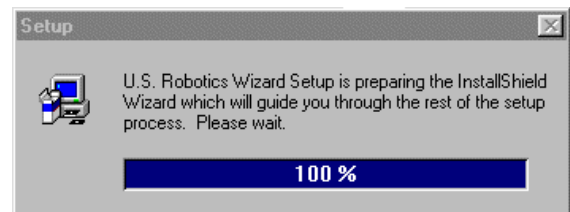
1. From the Windows 95 desktop, click the **Start** button and then click **Run**.



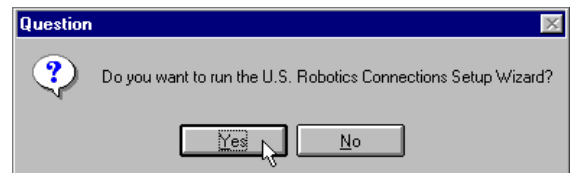
2. In the text box, type **D:\setup.exe**. (If your CD-ROM drive has a letter name other than **D**, type that letter in place of **D**.)



3. You will briefly see a screen that looks like this.



4. When you are asked if you wish to run the Setup Wizard, click **Yes**.



SOFTWARE INSTALLATION AND TESTING

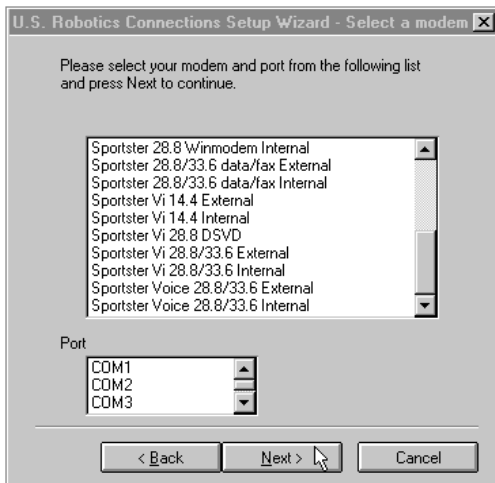
5. This is the Setup Wizard's "Welcome" screen. After reading the screen, click **Next**.



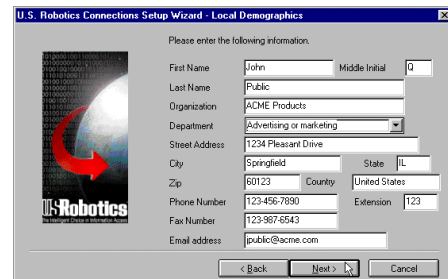
6. When you see the following screen, select your modem from the list, verify that the COM port setting is correct, and then click **Next**.

 **NOTE: Check the COM port setting against the setting you wrote down during the hardware installation.**

SOFTWARE INSTALLATION AND TESTING

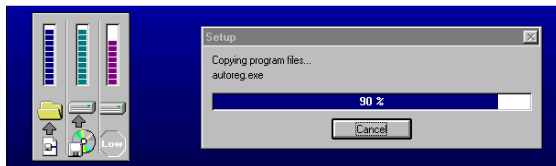


- When you see the next two screens, fill in the blank boxes with the appropriate information, using the **TAB** key to move between fields. Click **Next** on each screen when you have filled in all of the necessary information.

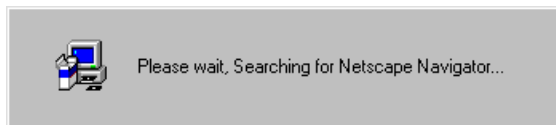


SOFTWARE INSTALLATION AND TESTING

8. You will see the following screen as the Setup Wizard creates a U.S. Robotics *Connections* program group.

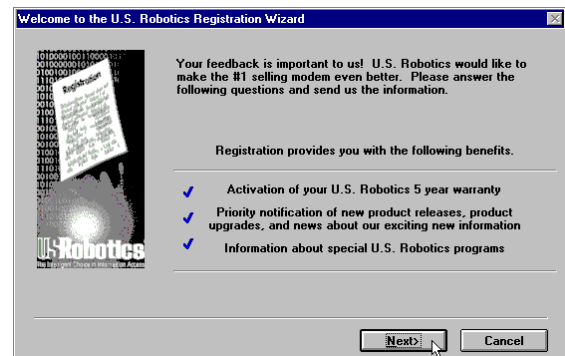


9. Next, the Setup Wizard looks for Netscape Navigator on your system.



If the Setup Wizard does not find Netscape Navigator, it searches for Microsoft's Internet Explorer. If the Setup Wizard does not find Internet Explorer, it will automatically install the application later in the Setup Wizard process.

10. The next screen introduces the U.S. Robotics Registration Wizard. After you read the screen, click **Next**. Read through each of the next two screens and verify or correct the information you typed earlier. Click **Next** on each screen to move on.



SOFTWARE INSTALLATION AND TESTING

User Information

Please verify the following information.

First Name: John Middle Initial: Q
Last Name: Public
Organization: ACME Products
Department: Advertising or marketing
Street Address: 1234 Pleasant Drive
City: Springfield State: IL
Zip: 60123 Country: United States
Phone Number: (123) 456-7890 Extension: 123
Fax Number: (123) 987-6543
Email address: jpublic@acme.com

<Back Next> Cancel

Product Information

Please fill out the following information about your product.

What kind of modem did you buy?
Date of Purchase
Where did you buy your modem?
Product Serial Number
You can find the serial number on the white sticker on the modem and on the outside of the box.
☒ Check here if you would like information on product upgrades from time to time to be sent to you.

<Back Next> Cancel

11. When you see this screen, you are ready to register your new modem. Click **By Modem**. We recommend you choose the **By Modem** option because it's a great way to verify that your Sportster modem is correctly installed.

You are now ready to register your U.S. Robotics modem

To take advantage of U.S. Robotics registration program, you may send this information to us by modem, fax, mail, or e-mail.

Please choose one of the following registration methods. For more information, select one method and click on the details box.

☒ By Modem
☐ By Fax
☐ By Mail
☐ By E-mail

Details...

<Back OK> Cancel

SOFTWARE INSTALLATION AND TESTING

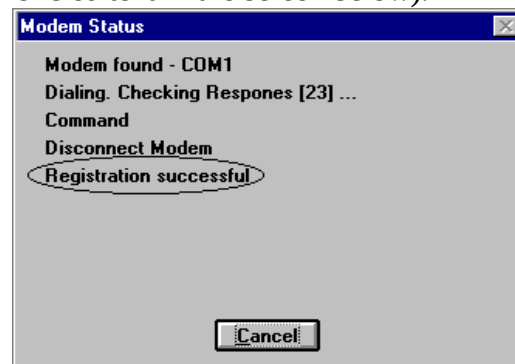
12. When you see this screen...



- If you need to dial a prefix (such as 9) to make a call outside your building, type the prefix before the 1 in the **Prefix** box and then click **Dial**. If the modem still does not dial the number properly, you may need to insert a comma between the prefix and the 1 to force the modem to pause after dialing the prefix.

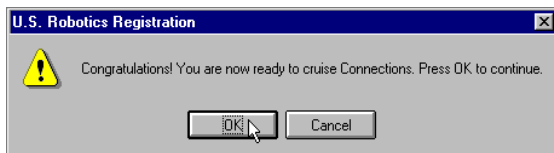
- If you do not need to dial a prefix, simply click **Dial**.

13. You will see a screen indicating that the registration information has been sent (see circled text in the screen below).



14. The next screen indicates that the Setup Wizard is finished. Click **OK** to launch the *Connections* CD-ROM.

SOFTWARE INSTALLATION AND TESTING

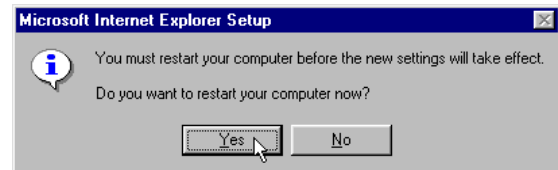


NOTE: If the Setup Wizard detected Internet Explorer on your system during the setup process, your computer will launch the *Connections* CD-ROM when you click **OK**. Continue at step 15.

If the Setup Wizard *DID NOT* detect Internet Explorer on your system earlier in the setup process, it will launch the Internet Explorer installation utility after you click **OK**. When you see the following screen, click **Install Internet Explorer** and follow the on-screen instructions to complete the installation of the software.



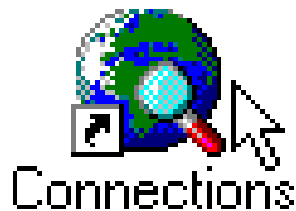
At the end of the installation process, you will see the following screen. Click **Yes**.



Windows 95 users: Windows will restart and the *Connections* CD will launch automatically upon restart.

SOFTWARE INSTALLATION AND TESTING

Windows 3.x users: Windows will restart. When your desktop reappears, you will see a **Connections** icon (shown below) in the **Connections** program group. Double-click the icon to launch the CD.



15. When the main *Connections* menu appears, click **Business & Productivity** on the left hand side of the screen (circled below).

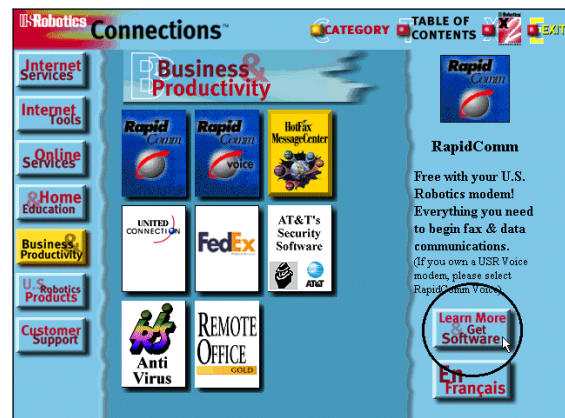


SOFTWARE INSTALLATION AND TESTING

16. On the **Business & Productivity** menu is a **RapidComm** button (circled in the following screen shot). RapidComm is the fax/data software you can use to send faxes directly from your desktop, transfer files electronically, or dial into a BBS. Click **RapidComm**.

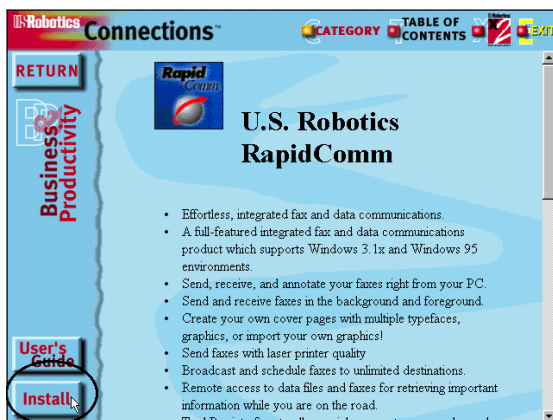


17. The next screen contains a menu on the right hand side of the screen. Included on the menu is a **Learn More & Get Software** button (circled in the screen image below). Click this button.



SOFTWARE INSTALLATION AND TESTING

18. The next screen displays information about RapidComm. When you are ready to install RapidComm, click **Install** (circled in the lower left hand corner of the following screen image).

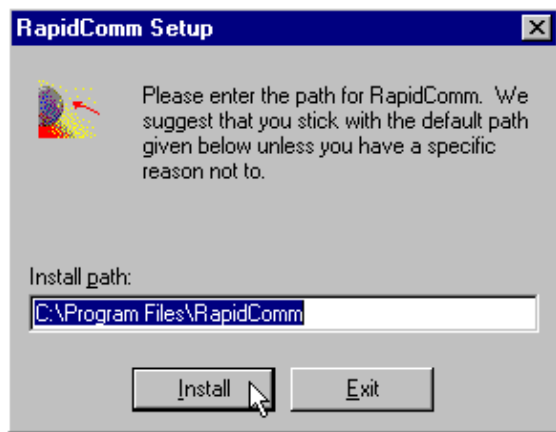


NOTE: If you are using Netscape Navigator to view your Connections CD-ROM, you may be asked to save the application's installer to your computer when you click the **Install** button. Note the installer's file name and its location on your computer. To run the installer or demo, find the file on your computer and double-click it. The installer will start.

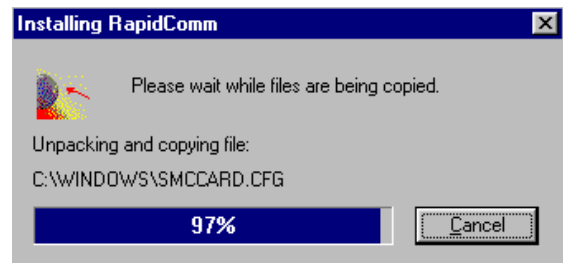
19. The next screen is the first of the “RapidComm Setup” screens. When this screen appears, make sure the location shown in the text box is where you would like the files copied to and then click **Install**. If you wish to copy the RapidComm files to a different directory,

SOFTWARE INSTALLATION AND TESTING

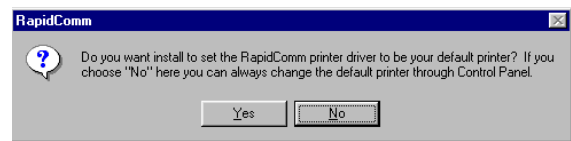
type that location in the text box before clicking **Install**.



20. You will see this screen as files are copied.



21. When you see this screen, click either **Yes** or **No**.

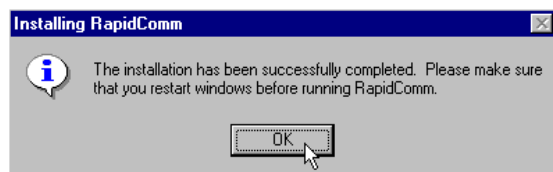


- If you click **Yes**, every document you print will be treated as a fax unless you change the printer selection in the program from which you are printing.
- If you click **No**, RapidComm will not be selected as the default printer. When you want to send a fax, you must select

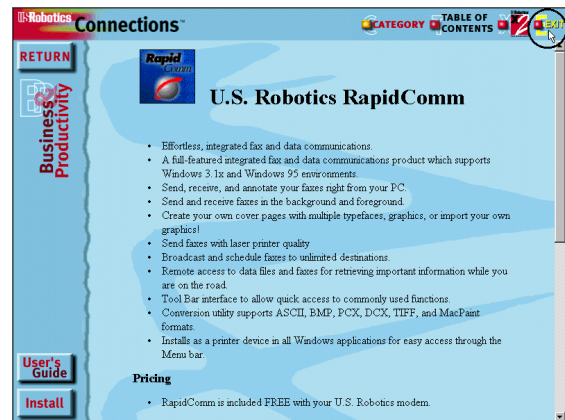
SOFTWARE INSTALLATION AND TESTING

RapidComm as the printer in the program from which you are printing.

22. This screen marks the end of the RapidComm installation. Click **OK**.

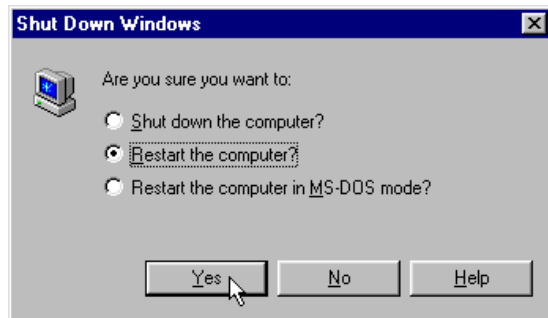


23. Click **OK** to return to *Connections*. Click **Exit** (circled in the following screen image).

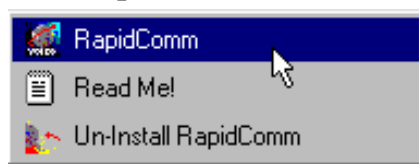


SOFTWARE INSTALLATION AND TESTING

24. Restart Windows by clicking the Windows **Start** button and pointing to **Shut Down**. Click **Restart the Computer?** on the *Shut Down Windows* screen. Then click **Yes**.



25. When your desktop appears, click Windows **Start**. Point to **Programs**. Then click **RapidComm**.

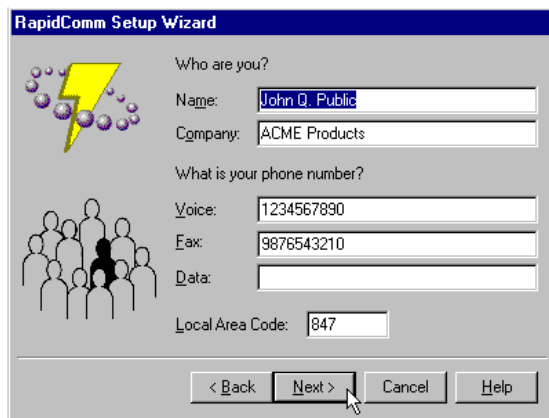


26. The following is the first RapidComm Setup Wizard screen. After you read the screen, click **Next**.



SOFTWARE INSTALLATION AND TESTING

27. When you see this screen, verify the information shown and then fill in the **Data** box (if you have a third phone number just for your modem) and the **Local Area Code** box. Then click **Next**.



The screenshot shows the 'RapidComm Setup Wizard' window. On the left is a graphic of a yellow lightning bolt with purple bubbles and a group of five stylized human figures. The main area contains the following fields:

- Who are you?**
 - Name: John Q. Public
 - Company: ACME Products
- What is your phone number?**
 - Voice: 1234567890
 - Fax: 9876543210
 - Data: (empty)
- Local Area Code:** 847

At the bottom are four buttons: '< Back', 'Next >', 'Cancel', and 'Help'. A mouse cursor is pointing at the 'Next >' button.

28. After you verify the information on this screen, click **Next**.



The screenshot shows the 'RapidComm Setup Wizard' window. On the left is a graphic of a yellow lightning bolt with purple bubbles and a globe showing North and South America. The main area contains the following fields:

- The following information is needed for sending faxes, creating cover pages, and placing calls...**
 - Address 1: 1234 Pleasant Drive
 - Address 2: (empty)
 - City: Springfield
 - State: IL
 - Country: United States
 - Zip/Post Code: 60123
 - E-Mail Address: jpublic@acme.com

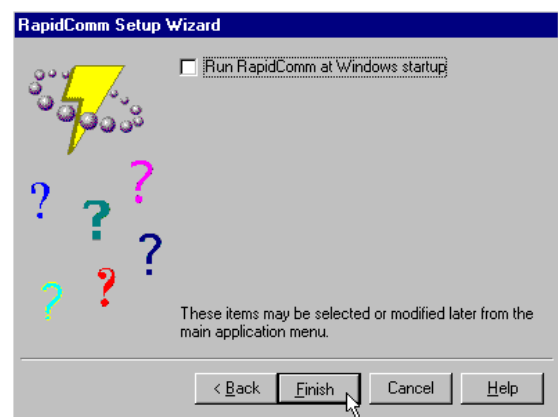
At the bottom are four buttons: '< Back', 'Next >', 'Cancel', and 'Help'. A mouse cursor is pointing at the 'Next >' button.

SOFTWARE INSTALLATION AND TESTING

29. Select the correct modem in the text box on the following screen. Then click **Next**.

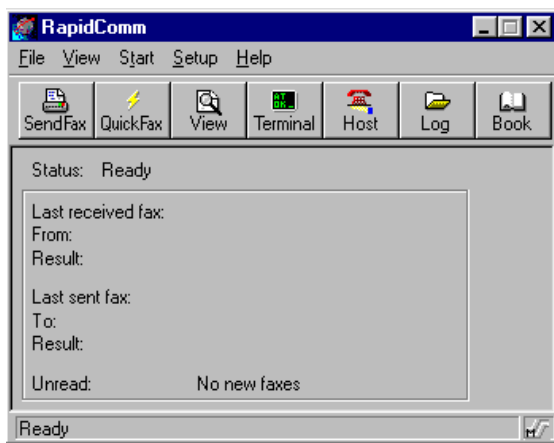


30. When you see this screen...



- If you want RapidComm to launch with every Windows startup, select the **Run RapidComm at Windows startup** check box. Then click **Finish**.
- If you do not want RapidComm to start every time you start Windows, simply click **Finish**.

31. You will see this screen.



This is the main RapidComm screen. To send your first fax, keep this screen open. (After this initial run, RapidComm will not need to be running in order to send a fax.)

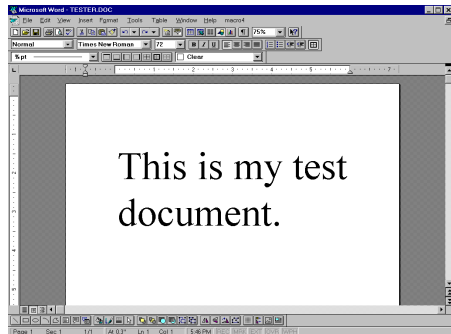
Sending Your First Fax

Using RapidComm, you can send and receive faxes directly from your computer and eliminate the need for an expensive fax machine. Once you learn the basics of sending faxes, you can learn more involved fax functions, such as sending documents to groups of numbers at assigned times and how to transfer data files. These more advanced functions are explained in the electronic RapidComm manual on your *Connections* CD. This chapter will walk you through sending your first fax.

1. Open an application in which you can create documents that you might want to fax (e.g. a word processing application). Create a document containing only a

SOFTWARE INSTALLATION AND TESTING

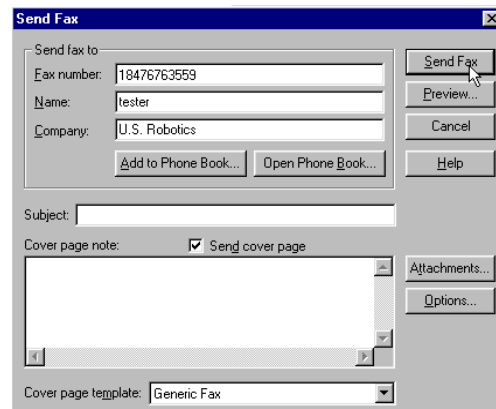
sentence or two. Name the document **tester**. Keep the document open.



2. From the **File** menu, click **Print**.
3. Select **RapidComm** as the printer. This can be changed in most Windows applications in the **Print** dialog box.
4. Click **OK** or **Print** (whichever button you click in your application to indicate that you are ready to print) in the **Print** dialog box.
5. When you see the following screen, fill in the necessary information in the text boxes.

For testing purposes, send your “tester” document to the U.S. Robotics fax number, 847-676-3559.

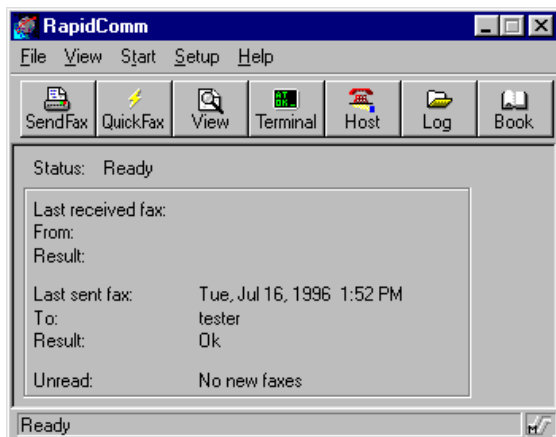
 **NOTE: If you need to enter a prefix (such as 9) to dial a number outside your building, type the prefix before the fax number in the Fax number field.**



SOFTWARE INSTALLATION AND TESTING

When you finish, click **Send Fax**.

6. You will see a series of screens as the fax is being transmitted. When the fax has been successfully transmitted, you will see a screen that looks like this.



Congratulations —you are now ready to start using your Sportster modem!

Go to the electronic RapidComm manual on your *Connections* CD for more detailed instructions on sending faxes and other things you can do using your Sportster modem and RapidComm.

INSTALLING OTHER FAX/DATA SOFTWARE

You do not have to use the RapidComm fax/data software included with your new Sportster modem. Your modem was designed for and tested using a wide range of communications software packages. This section will guide you through some of the details you may need to know when installing other communications software packages.

Type of Modem

Most communications software programs will ask you to select the type of modem you are using. Select a *U.S. Robotics Sportster high speed modem*. If that selection is not listed, pick *Courier Dual Standard, V.32bis*, or *V.34*.

D KEY POINT: Refer to the manual that came with your software for its installation instructions. The software's installation program will ask you questions about the modem you are using.

Initialization String

For hardware flow control, a fixed serial port rate and full result codes type **AT&F1** and then press **ENTER**.

If you must use software flow control, type **AT&F2** and then press **ENTER**.

Flow Control

Σ For hardware flow control (highly recommended), select **RTS/CTS**.

- For software flow control, select **XON/XOFF**.



NOTE: You may need to disable the flow control (hardware or software) that you are not using.

UART (External Modems)

If you are running Windows 3.x or you have upgraded your system from Windows 3.x to Windows 95, you can run MSD to determine your UART setting. In DOS, type **MSD** at the Windows directory prompt and then press **ENTER**. Follow the on-screen instructions to access the COM port settings panel. In this panel you should find the UART chip used. Match the UART type listed in MSD with the serial rate listed in the chart below. Select this

serial rate in any communications software you use.

<u>If this is your UART...</u>	<u>Select this serial rate</u>
16550*	115.2 or 57.6 kbps
16450	38.4 kbps
8250	19.2 kbps

*All Sportster internal modems have a 16550 UART.



NOTE: DO NOT select a 28,800, 14,400, or 12,000 bps serial port rate, if offered. Your modem will NOT work correctly with any of these settings. Fix or lock the serial port (baud) rate (if it's referred to as autobaud, select OFF).

USING MODEM STATION

What Does Modem Station Do?

- ◆ Modem Station provides a simple to use interface that takes the pain out of communicating with your modem.
- ◆ Modem Station allows you to point and click your way through configuration.
- ◆ Modem Station can automatically detect your modem and provide you with all the technical information you need, whenever you need it!

Why Modem Station?

- ◆ Modem commands can be confusing and difficult to memorize.
- ◆ Communications software often needs technical information about your modem.

- ◆ You may want to “tweak” your modem for optimum performance.
- ◆ You’d rather be surfing the Internet.

Installing Modem Station

If you did not install Modem Station when you first installed the *Connections* CD, please follow these instructions.

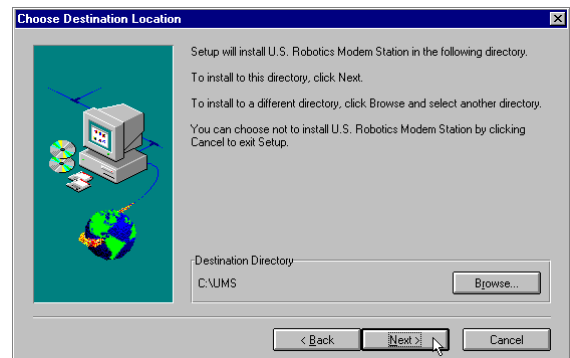
1. Insert the *Connections* CD into your CD-ROM drive.
2. Double-click the **My Computer** icon on your desktop.
3. Double-click the **CD-ROM** icon.
4. Double-click the **USR Tools** folder.
5. Double-click the **umssetup** icon.

USING MODEM STATION

6. You will be asked whether you wish to install Modem Station. Click **Yes**.
7. Wait a few moments for the Installation Wizard to load.
8. After reading the information on the “Welcome” screen, click **Next**.

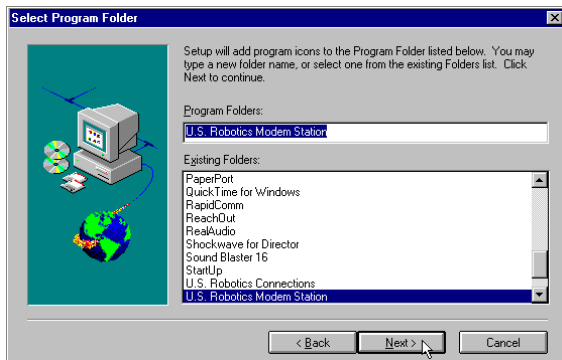


9. When you see this screen, click **Next** to accept the default directory or click **Browse** to change directories.

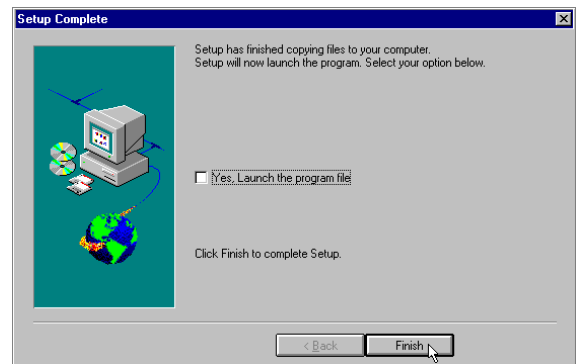


USING MODEM STATION

10. Click **Next** on the following screen to accept the default program folder. You can place Modem Station in an existing folder by selecting one from the list.



11. When you see the following screen, click **Finish** to complete the installation.

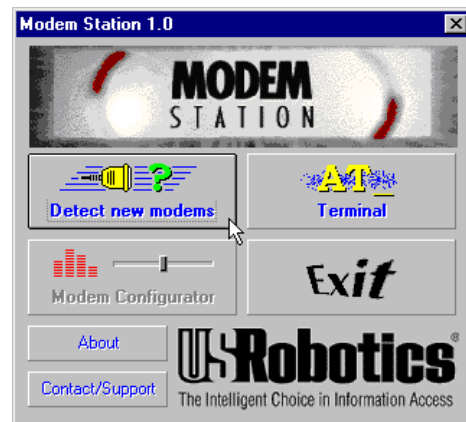


If this is the first time you've installed Modem Station, you may be asked to restart your computer.

USING MODEM STATION

Starting Modem Station

1. If you didn't start Modem Station from the Setup program, please start it now.
2. Click Windows **Start** button and then point to **Programs**. Click **U.S. Robotics Modem Station** (or the folder you selected during installation).
3. Double-click the **Modem Station** icon. This brings up the main menu.



USING MODEM STATION

The main menu gives you direct access to the following options:

- ◆ DETECT NEW MODEMS
- ◆ TERMINAL
- ◆ MODEM CONFIGURATOR
- ◆ ABOUT
- ◆ CONTACT/SUPPORT

DETECT NEW MODEMS

This option detects US Robotics modems installed on your system and shows what COM port they are using. Click this option if you are running Modem Station for the first time, if you are changing modems, or if you simply need to know what port your modem is using.

TERMINAL

Terminal allows you to send commands directly to your modem and displays the

responses. You can use Terminal to dial up BBSs. In addition, you can configure your modem using Terminal. However, it is much easier to use the Modem Configurator.

MODEM CONFIGURATOR

Modem Configurator provides an easy-to-use interface for entering hard to remember commands. Use Modem Configurator for troubleshooting, initial configuration, and tuning your modem for optimum performance. Using the options available in Modem Configurator, you can control nearly every aspect of your modem's performance. We will discuss Modem Configurator's options in more detail in later sections.

ABOUT

The About option provides copyright and version information.

USING MODEM STATION

CONTACT/SUPPORT

This option details how to get in touch with U.S. Robotics.

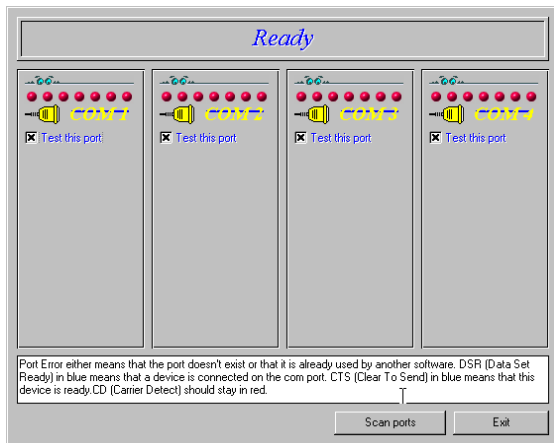


TIP: For your convenience, we provide many on-line support avenues. For specific questions, our fax-on-demand service is a good place to start. You can download FAQs, software, and help files from our Web sites and BBS, or receive individualized support via support@usr.com. Type 0000 (4 zeroes) in the subject line of your e-mail.

USING MODEM STATION

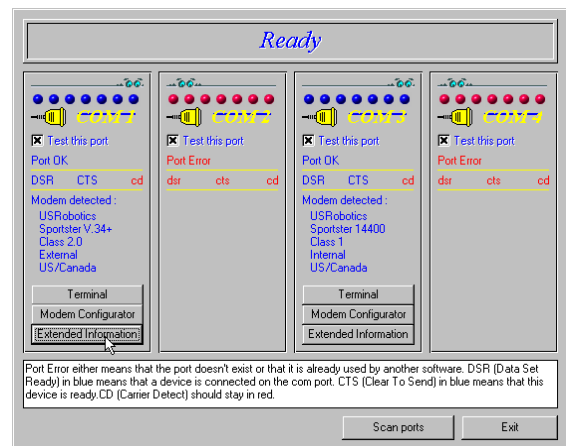
Using Detect New Modems

1. Click **Detect New Modems** to bring up the following screen.



The screen consists of four columns, one for each possible COM port on a PC. You can scan a specific port(s) by selecting the checkbox for that port.

2. Click **Scan** to have Modem Station check for installed modems. This may take a few moments.
3. When the scan finishes, you will see the following display. Your display may differ depending on the type and number of modems installed.



USING MODEM STATION

If your modem is installed and configured correctly, Modem Station will find the modem and display make and model information under the assigned port. All currently active ports should display “Port OK” under the heading. If a port displays a “Port Error”, it usually means that the port is disabled in system setup.



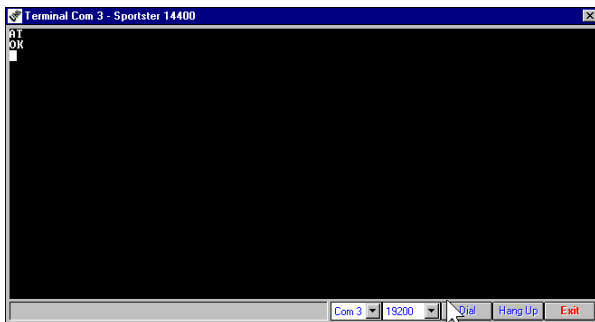
NOTE: Different systems and BIOSes use different methods of disabling COM ports. As a result, we cannot provide support for enabling COM ports. Please refer to your system’s documentation or contact the manufacturer of your system for further information.

If you look at the information for the port your modem is using, you will see three buttons. These allow you to access Terminal and Modem Configurator without going back to the main menu. Extended Information provides detailed information about your modem, previous connections, firmware dates, etc. This next section details using the Terminal option.

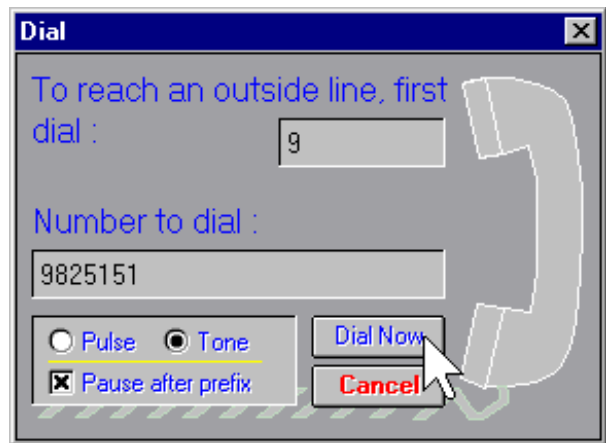
Using Terminal

You can access Terminal from either the **Main** menu or the *Detect New Modems* screen. Clicking **Terminal** brings up the *Terminal* window.

USING MODEM STATION



In addition to allowing direct entry of modem commands, the *Terminal* window also allows you to dial into Bulletin Boards, listing services, and other online services.



USING MODEM STATION



NOTE: Modem Station's Terminal window is provided primarily for troubleshooting convenience. If you frequently use BBSs, you will probably want to use a separate, full-featured Terminal program such as that provided in our RapidComm software.

On the lower part of the Terminal screen, you will see the COM port your modem is currently using. To select another modem, simply click on the arrow and select that modem's assigned port.

To the right of the port settings are the port speed settings. Port speed is the speed at which your computer sends data to the

modem. We will discuss port speed settings in detail later in this section.

Terminal includes a basic auto dialer.

To have Terminal dial a number for you, click **Dial** to bring up the "Dial" screen.

You need to tell the Dialer a few things about your phone system, such as whether it uses tone or pulse dialing, what digit, if any, you need to dial to get an outside line, and whether the dialer should wait between dialing that digit and the rest of the number. Once you provide this information, simply enter the phone number as if you were dialing a telephone. Click **Dial Now** to dial the number.

You can end a call by clicking **Hang Up** at the bottom of the screen.

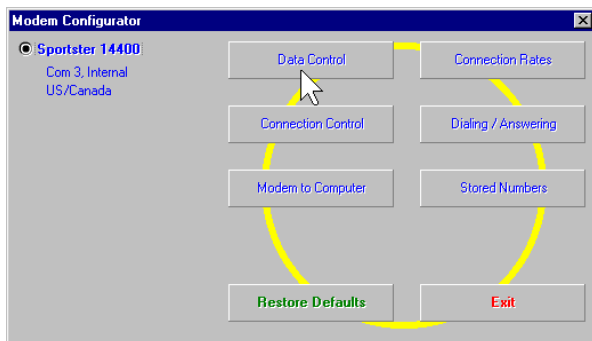
USING MODEM STATION

When you are finished using Terminal, click **Exit** to return to the screen you accessed it from.

Using Modem Configurator

You can access Modem Configurator from either the **Main** menu or the *Detect New Modems* screen.

Click **Modem Configurator** to bring up this menu.

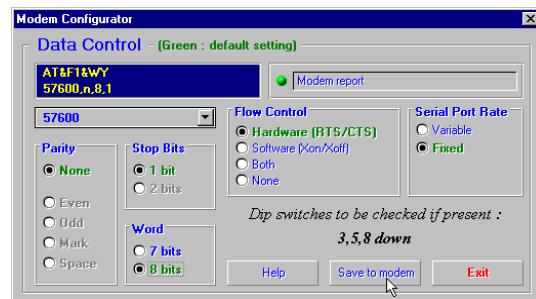


The **Modem Configurator** menu gives you access to the following options:

- ◆ DATA CONTROL
- ◆ CONNECTION CONTROL
- ◆ MODEM TO COMPUTER
- ◆ CONNECTION RATES
- ◆ DIALING/ANSWERING
- ◆ STORED NUMBERS
- ◆ RESTORE DEFAULTS
- ◆ EXIT

Data Control

This is the “Data Control” screen.



USING MODEM STATION

The “Data Control” screen allows you to assign the following basic communications settings:

- ◆ PORT SPEED
- ◆ PARITY
- ◆ STOP BITS
- ◆ WORD
- ◆ FLOW CONTROL
- ◆ SERIAL PORT RATE

For information on using these settings, please refer to the “Glossary” at the back of this manual.

Click **Help** for quick definitions of the terminology used in this screen.

In the upper left-hand corner of the screen, you will see the data control commands currently in use.

Once you have entered the Data Control settings, click **Save to Modem**. This stores the settings so that you do not have to re-enter them.

This screen also displays the default DIP switch settings.



TECHNICAL STUFF: DIP switches are tiny switches that control a few basic functions on some external modems. On modems without DIP switches, these functions are handled by modem commands.

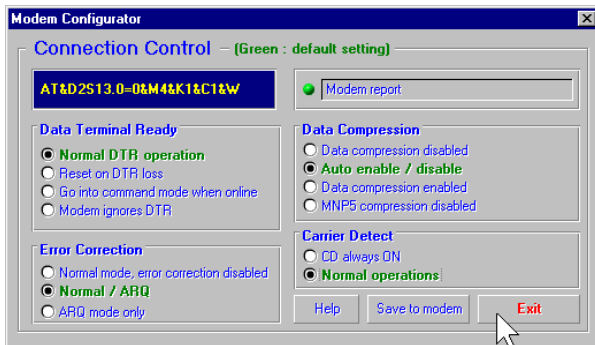
Click **Exit** to return to the **Modem Configurator** menu.

USING MODEM STATION

CONNECTION CONTROL

This is where you adjust your modem's connection and transmission settings.

Click **Connection Control** to bring up this screen.



In the upper left hand corner of this screen, you will find the current Connection Control settings.

Use the “Connections Control” screen to configure the following settings:

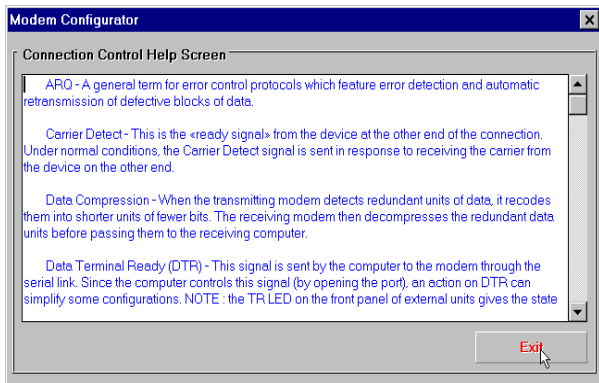
- ◆ DATA TERMINAL READY (DTR)
- ◆ ERROR CORRECTION
- ◆ DATA COMPRESSION
- ◆ CARRIER DETECT

For detailed information about these settings, refer to the “Glossary” or the “Technical Quick Reference” sections of this manual.

 **TIP: On external modems, receiving a Data Terminal Ready signal causes the TR light to light up.**

Click **Help** for quick definitions of terminology used in this screen.

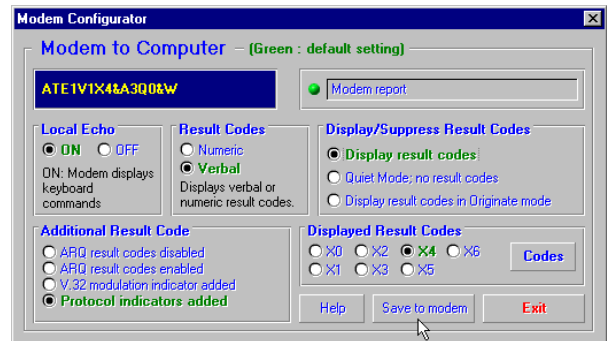
USING MODEM STATION



Once you have configured your Connection Control settings, click **Save to Modem** to save your settings

Using Modem to Computer

These settings control how your modem and computer communicate with each other. They control what you see on your terminal screen and how results are displayed.



If you look in the upper left hand corner of the display, you will see the commands currently in use.

The “Modem to Computer” screen allows you to configure the following settings:

- ◆ LOCAL ECHO
- ◆ RESULT CODES
- ◆ RESULT CODE PREFERENCES

USING MODEM STATION



TIP: If you type ATDT and see 'AATTDDTT' on your screen, it is possible that both your software and modem have Local Echo set to 'ON'. Turn Local Echo 'OFF' on EITHER the modem or the software to solve this problem.

For details on using the commands in this screen, refer to the "Glossary" or "Technical Quick Reference" sections of this manual.

Click **Help** to see quick definitions of terminology used in this screen.

This screen also includes a chart of the ways result codes can be displayed. Click **Codes** to view a chart of the display options.

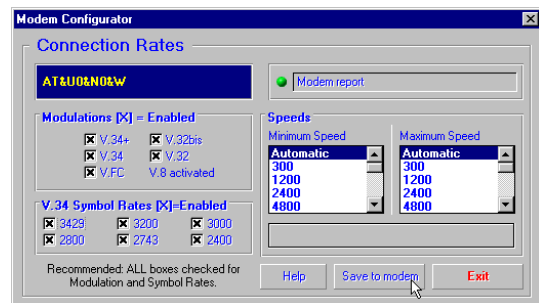
Once you configure your settings, click **Save to Modem** to save your choices.

CONNECTION RATES

The "Connection Rates" screen allows you to configure modem speeds and protocols.



WARNING! Use caution when changing connection settings. Improper settings may cause your modem to function incorrectly, disconnect, or fail to connect at all.



USING MODEM STATION

In the upper left hand corner of the screen you will see the current connection commands.

This screen allows you to configure the following settings:

- ◆ MODULATIONS
- ◆ V.34 SYMBOL RATES
- ◆ SPEEDS

Again, once you have selected your settings, click **Save to Modem** to save them.

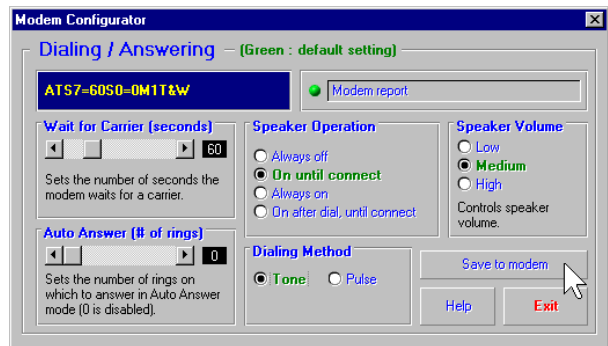
When you are finished, click **Exit** to return to the **Modem Configurator** menu.

Please refer to the main body of the manual and the “Glossary” for detailed information about the terminology and settings used in this screen.

Click **Help** for quick definitions of terminology used in this screen.

DIALING/ANSWERING

The next screen allows you to adjust how your modem initiates and receives calls.



Using this screen, you can configure the following dial settings:

- ◆ WAIT FOR CARRIER
- ◆ AUTO-ANSWER # OF RINGS
- ◆ SPEAKER OPERATION
- ◆ DIALING METHOD
- ◆ SPEAKER VOLUME

USING MODEM STATION



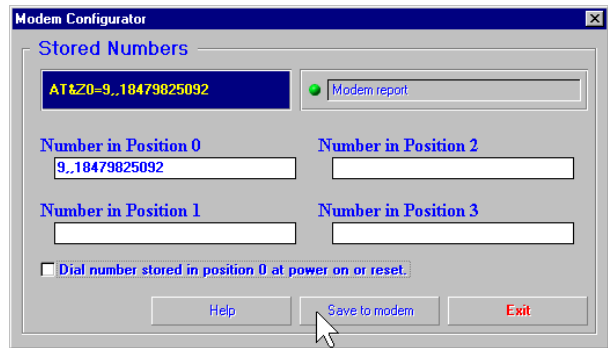
TECHNICAL STUFF:

“Negotiation” is the noise we hear when two modems connect. The modems exchange information about their respective protocols and configurations. Once they agree on the fastest protocol they can both handle, transmission begins.

STORED NUMBERS

The “Stored Numbers” screen displays the phone numbers currently stored in your modem’s memory.

Use this screen to edit or add numbers stored in the modem’s memory.



To store a number in your modem’s memory, simply click in one of the entry boxes. Type in the phone number exactly as you would dial it. NOTE: Position Zero has a special feature. You can set your modem to automatically dial this number when your computer is turned on or when it is reset.

This is very useful if you are using your modem with a “dumb terminal” or know that you need to connect to a specific bulletin board or listing service.

USING MODEM STATION

You can change stored numbers by highlighting them and then typing the new numbers in their place. Once you store your numbers, you can dial them by entering this command from Terminal Mode:

ATDS0, ATDS1, ATDS2, etc.

Your U.S. Robotics modem comes with one phone number already stored in Position 0. If you haven't changed the default, typing **ATDS0** will automatically dial the U.S. Robotics BBS. You will want to change this if you are using a dumb terminal or online listing service, of course.

Once you enter the numbers you wish to store, click **Save to Modem** to store them. Click **Exit** to return to the **Main** menu.

The last option, Restore Defaults, resets your modem to factory specifications. This option is

available from many of the screens within Modem Station. When you click **Restore Defaults**, you will see a confirmation screen asking if you are sure you want to restore defaults. Click **OK** to proceed.



TIP: Restore Defaults will set your modem back to factory specifications. It is a good place to start when troubleshooting.

Using the Extended Information Screens

The "Extended Information" screens provide important and useful information about your modem. Unfortunately, some of this information is very complex.

USING MODEM STATION

We devote a separate section to the Extended Information screens so that we can explain what you'll see (and why it is important to you) as fully as possible.

There are a series of commands used to obtain detailed information from U.S. Robotics modems. Extended Information provides a convenient way to get that important information without memorizing the commands.

From the "Detect New Modems" screen, click **Extended Information**.

The "Extended Information" screen provides access to the following information about your modem:

- ◆ ROM CHECKSUM
- ◆ PRODUCT
- ◆ ACTIVE PROFILE
- ◆ STORED PROFILE

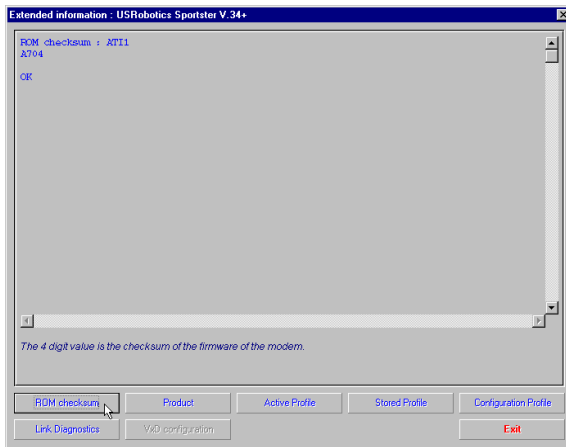
- ◆ CONFIGURATION PROFILE
- ◆ LINK DIAGNOSTICS
- ◆ VxD CONFIGURATION
- ◆ DIAL/SECURITY

Winmodem users will have access to VxD information via these screens.

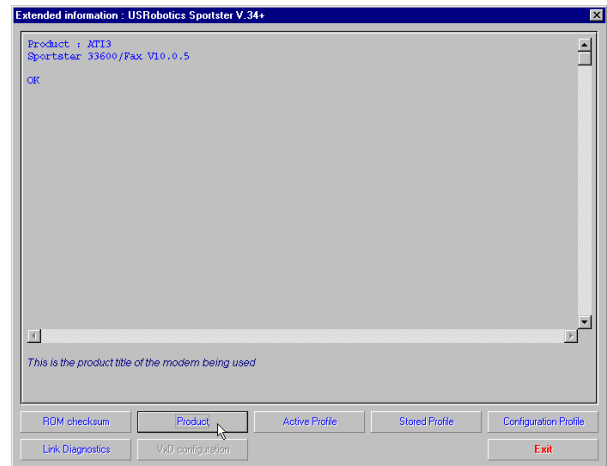
Courier users will have access to Dial/Security information.

Click **ROM Checksum** to bring up the following screen.

USING MODEM STATION



Technicians use the ROM Checksum to verify information stored in the modem's Read Only Memory. You can check the ROM Checksum from any Terminal program by typing **ATI1** and pressing **ENTER**. For information about the specific modem you are using, click **Product** to bring up the next screen.

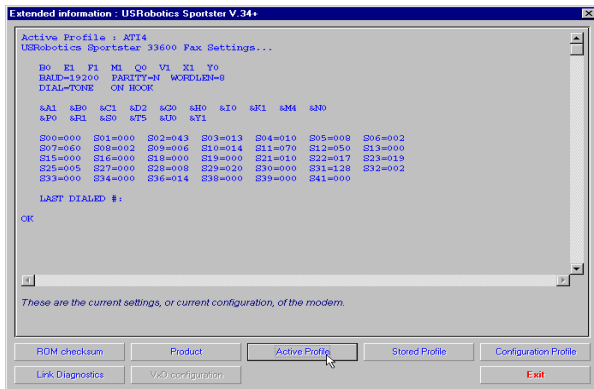


The “Product” screen displays the make and model of your modem. It is the equivalent of the ATI3 command.

Your modem is able to store two configurations or “profiles.” Only one can be active at any time.

USING MODEM STATION

To see information about the profile in use, click **Active Profile** to bring up the next screen.



This screen contains information about your modem's current configuration. Starting from the top, you will see the make and model of your modem. Directly below that you will see basic commands currently in use. Below that are the current connection settings.

The two lines below the connection settings are the advanced commands currently in use. Below them is a display of the contents of the 'S-Registers' for your modem. These registers are special programmable areas of your modem's memory. They are used to store commands that are too complex to be handled by the standard (or 'AT') commands.

Just below that you will see the last number dialed. You can also see this information by typing **ATI4** in a Terminal program and pressing **ENTER**.

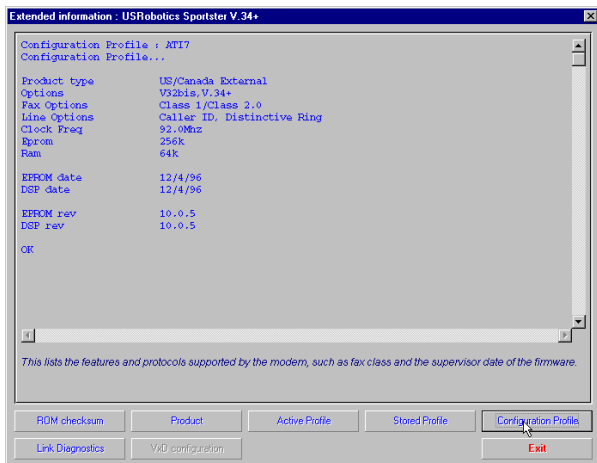
The next option displays the "Stored Profile" screen. This screen shows the configuration stored in your modem's NVRAM (special programmable memory). Note that any stored phone numbers are displayed on this screen as well. This is the equivalent of the ATI5 command. The information displayed is in the

USING MODEM STATION

same format as the “Active Profile” screen with the addition of stored phone numbers.

The next two screens contain information that our technical support representatives may need if you request support.

Click **Configuration** to bring up the first of these screens.



The “Configuration” screen displays the following information about your modem.

- *Product Type* displays the information relating to the make and model of your modem.
- *Options* displays the protocols available to your modem.
- *Fax Options* displays your fax compatibility.
- *Clock Frequency* displays the speed of the tiny “clock” that controls the timing of operations within the modem.
- *EPROM* displays how much information can be stored in the EPROM (or Supervisor) chip.
- *RAM* displays how much memory your modem has for processing commands and internal functions.
- *Supervisor Date* (or EPROM Date) displays the version date of the ‘firmware’

USING MODEM STATION

stored in the Supervisor chip. (The Supervisor chip contains the special software used to control your modem's functions.) If a support representative asks you for your Supervisor or EPROM date, look here.

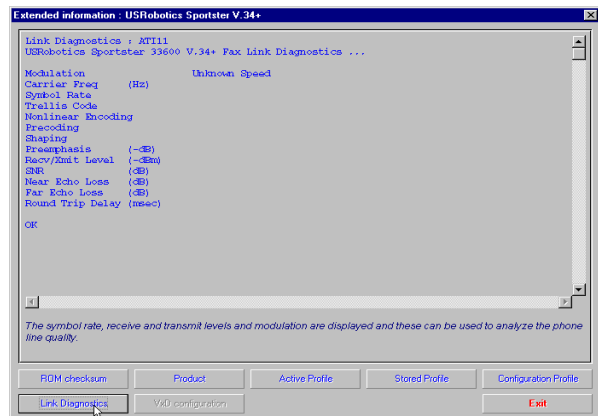
- *DSP Date* is the date of the 'firmware' that controls the DSP in your modem.
- *Supervisor and DSP rev* (or revisions) are the equivalent of software version numbers.



TECHNICAL STUFF: The Supervisor chip contains the commands that control your modem. The DSP controls execution of those commands. The Supervisor chip needs orders from the DSP before it can carry out a command.

- *DAA Country* displays the countries your modem is designed for. This is important because phone systems and telecommunications laws vary from country to country.

Click **Link Diagnostics** to bring up the "Link Diagnostics" screen.



USING MODEM STATION

The “Link Diagnostics” screen displays statistics about your last connection. This screen is most often used as an aid in diagnosing connection failures, but it also can provide information about connection speeds and phone line conditions.

- *Modulation* displays the speed and type of connection.
- *Carrier Frequency* displays the electrical frequency of the carrier signal.
- *Symbol Rate* displays the speed of the transmission.
- *Trellis Code, Nonlinear Encoding, Shaping, and Precoding* all refer to methods of handling high speed data transmission.
- The items followed by *(dB)*, *(-dB)*, *(-dBm)*, and *(msec)* refer to variations in the modulation tones that actually carry the information.

The rest of the screen contains information about CRC errors, Block Errors (Blers), resent data, and other data. This information is very important if you experience problems transmitting or receiving data or if you suspect problems with your phone service.

Only our U.S. Robotics Winmodems use VxDs, or Virtual Device Drivers. The Winmodem uses special software (called a ‘driver’) to manage many of the functions handled by hardware in our other modems. If you are using a Winmodem, the Extended Information menu will offer you the option of clicking on VxD to view information on the Winmodem driver. Please refer to your Winmodem manual for detailed information on the Winmodem drivers.

Courier users will see a display of current Dialback/Security settings. Refer to your

USING MODEM STATION

Courier documentation for instructions on configuring these special features.

General Installation Guidelines

Here are a few tips that may make setting up your new modem as simple as possible:

- ◆ Make certain you have everything you need before you get started.
- ◆ Take a few minutes to read through the manual(s) that come with your modem.
- ◆ Have a small jar or cup handy to hold screws and jumper shunts.
- ◆ Choose a time when you won't be interrupted.
- ◆ Check in advance to see what IRQs you have available.
- ◆ Set the jumper shunts on your modem (if necessary) before installation.
- ◆ If you are doing a Plug and Play installation with Windows 95, remember to remove all the jumper shunts from your jumper pins first.
- ◆ Before installing your modem, write the serial number in the space provided on the first page of this manual.
- ◆ Once you've installed your modem, make a note of the COM port and IRQ settings you are using on the first page of this manual. You may need this information to configure your communications software.

We hope that you enjoy exploring the world of information and entertainment your new Sportster modem will open up for you and that Modem Station will make using and

USING MODEM STATION

configuring your new modem as simple as possible.

Remember: If you have any questions or problems, feel free to visit one of our many online forums.

For answers to specific questions, be sure to visit our World Wide Web support site at **www.usr.com/home/online/main.htm**, or try our BBS (847-982-5092).

U.S. ROBOTICS MODEM UPDATE WIZARD

Your *Connections* CD-ROM includes the U.S. Robotics Modem Update Wizard. This software is designed to quickly update your modem, whether to a newer version of its current code or to U.S. Robotics' revolutionary new x2 technology. You can update your modem in any of the following 3 ways:

33.6 K	to	33.6 K (update)	= FREE
33.6 K	to	56 K (upgrade)	= \$FEE*
56 K	to	56 K (update)	= FREE

* Fee for upgrades is subject to change.

 **NOTE:** You can obtain this software from our BBS (847-982-5092) or from our World Wide Web page (www.usr.com) if you do not have a *Connections* CD-ROM.



NOTE: Complete the instructions in the Software Installation and Testing section of this manual before installing the Modem Update Wizard.



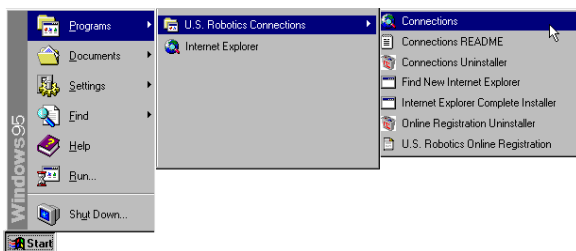
NOTE: These instructions pertain to Windows 3.x and Windows 95. However, only Windows 95 screen shots are shown unless the process for Windows 3.x users differ significantly.

Installation

1. Insert the *Connections* CD-ROM into your CD-ROM drive.

U.S. ROBOTICS MODEM UPDATE WIZARD

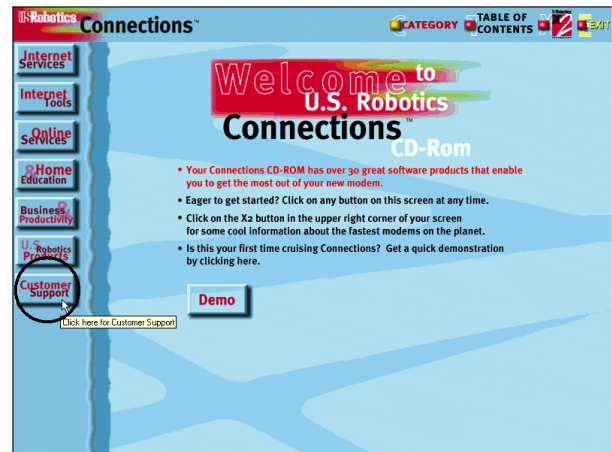
- Click the Windows **Start** menu and point to **Programs**. Point to **U.S. Robotics Connections**. Then click **Connections**.



 **NOTE:** Windows 3.x users launch the CD-ROM by clicking the Connections icon in the Connections program group.



- From the main **Connections** menu, click **Customer Support** (circled in the following screen shot).



U.S. ROBOTICS MODEM UPDATE WIZARD

4. When the **Customer Support** menu appears, click **Modem Update Wizard** (circled in the following screen shot).

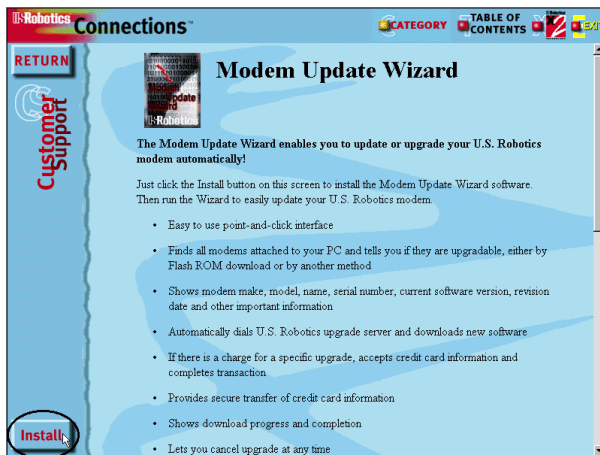


5. Next, click **Learn More & Get Software** (circled in the following screen shot).

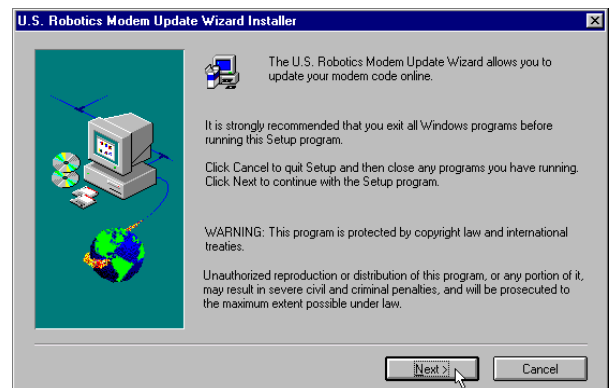


U.S. ROBOTICS MODEM UPDATE WIZARD

6. In the following screen, click **Install** (circled below).

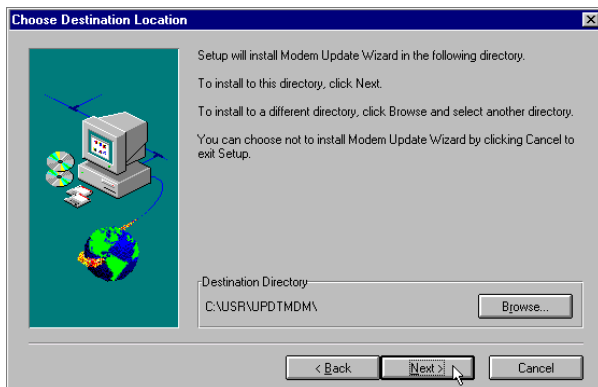


7. This launches the U.S. Robotics Modem Update Wizard Installer. The screen that follows is the welcome screen for the installer. After you read this screen, click **Next**.

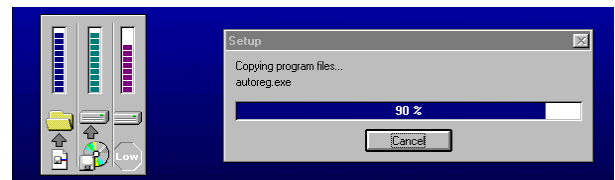


U.S. ROBOTICS MODEM UPDATE WIZARD

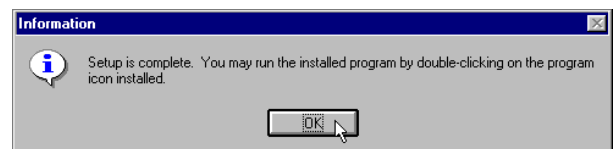
8. The next screen indicates where the installer will store the files for the Update Wizard. If the indicated location is acceptable, click **Next**. Otherwise, click **Browse** to find an acceptable directory.



9. You will see the following screen for a moment as the installer creates the Modem Update Wizard program group.



10. This screen indicates that the setup is complete. Click **OK**.



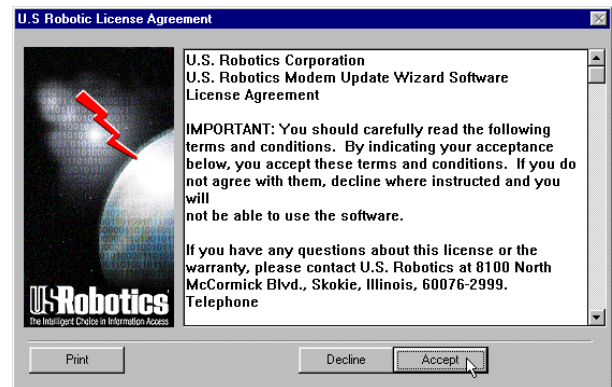
U.S. ROBOTICS MODEM UPDATE WIZARD

Operation

1. Click the Windows **Start** button. Point to **Programs**. Then point to **U.S. Robotics Modem Update Wizard**. Finally, point to the **Modem Update Wizard** selection.
2. After you read the "Welcome" screen, click **Next**.

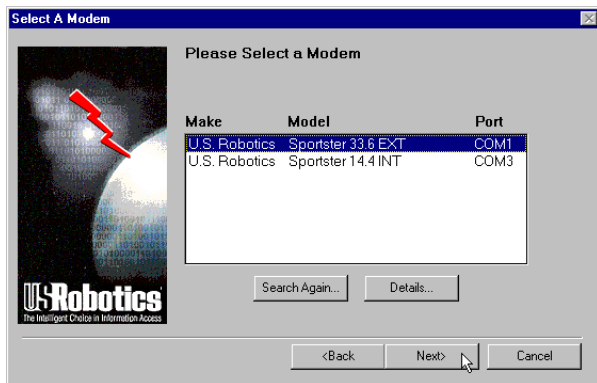


3. This brings up the license agreement. After reading the license agreement, click **Accept** to continue.



U.S. ROBOTICS MODEM UPDATE WIZARD

4. The software will now detect any modems installed on your system. When the detection phase finishes, you will see the following screen (the modems shown on your screen may differ). Click the modem you wish to upgrade and then click **Next** to continue.



5. The program will now determine if the selected modem is software upgradable.

Non-Upgradable Modems

If your modem **IS NOT** software upgradable, the program will display the following screen.



- **Click U.S. Robotics** if your modem is a U.S. Robotics modem to learn more about your upgrade options.

U.S. ROBOTICS MODEM UPDATE WIZARD

- **Click Other** to learn more about how to purchase a U.S. Robotics x2 modem.

Upgradable Modems

If the modem **IS** software upgradable, you will see the following screen.



Click **Dial U.S. Robotics** (which should be the default selection) in the pull-down menu. Click **Upgrade to x2 Technology** if you wish

to upgrade a 33.6 kbps modem to 56 kbps code or click **Update my modem's code** to update your modem with the most recent version of its original code. Then click **Next**.

6. The next screen is the order information screen. Users who are purchasing a 56K upgrade will see the first screen on the next page. It includes a credit card information query. The second screen appears to users who are updating their modem's code. Fill in the screen with the appropriate information and then click **Next**.

U.S. ROBOTICS MODEM UPDATE WIZARD

Users Purchasing 56K Upgrades

Fill out the Order Form

Please fill out the purchase form below. This information will be used to order an upgrade should you choose to buy one.

First Name M.I.
Last Name
Address
City
Province Zip
Country
Credit Card
Account # Exp. Date

<Back Next> Cancel

Users Updating Their Modem's Code

Fill out the Order Form

Please fill out the order form below.

First Name M.I.
Last Name
Address
City
State Zip
Country
Phone

<Back Next> Cancel

7. This screen dials the U.S. Robotics Update Server. Click **Tone** or **Pulse** dialing as necessary. If you normally dial a prefix (such as 9) to dial out on your phone line, type that prefix before the 1 in the **Prefix** box. Then click **Dial**.

US Robotics' Server

Select a Port

Dialing Options

☒ Tone ☐ Pulse

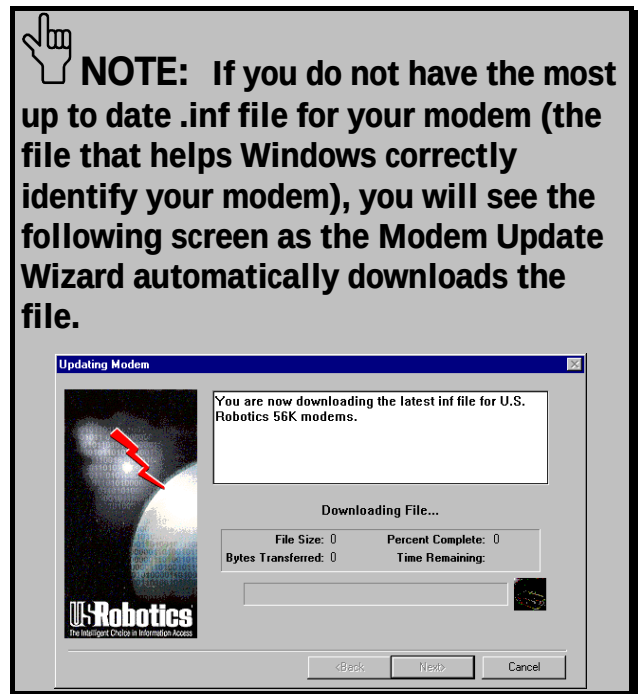
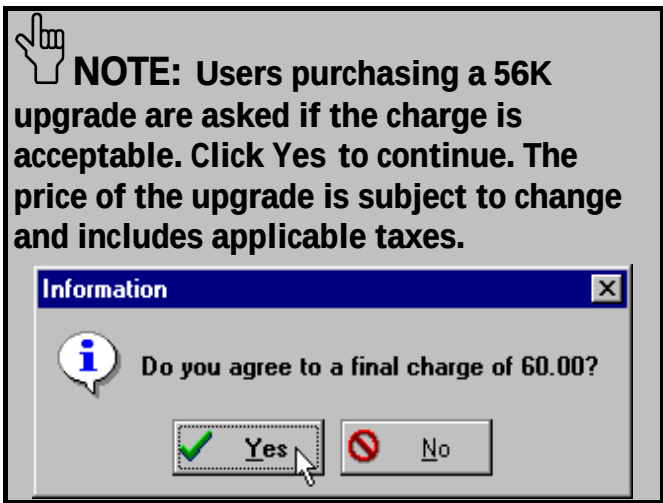
Calling From

Prefix Number

Dial Cancel

U.S. ROBOTICS MODEM UPDATE WIZARD

8. Once connected, the server will compare available upgrades with the current code in your modem. If they differ, the server will automatically initiate the download procedure.



U.S. ROBOTICS MODEM UPDATE WIZARD

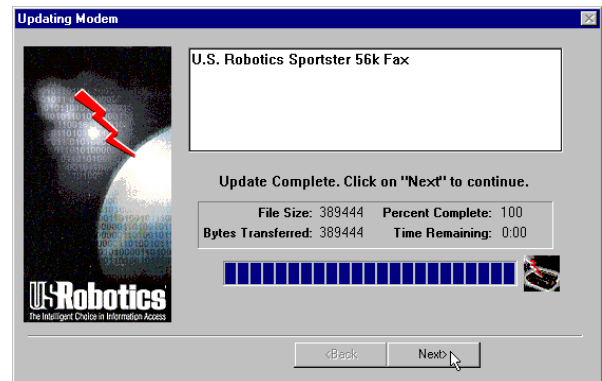
9. You will see this screen as the Wizard downloads the file for your modem.



10. When the download is complete, the modem disconnects from the server. The words "Downloading file..." change to "Upgrading ..." or "Updating ...", depending on whether you are upgrading or updating your modem. A new progress bar appears. This indicates that your modem's code is being updated.

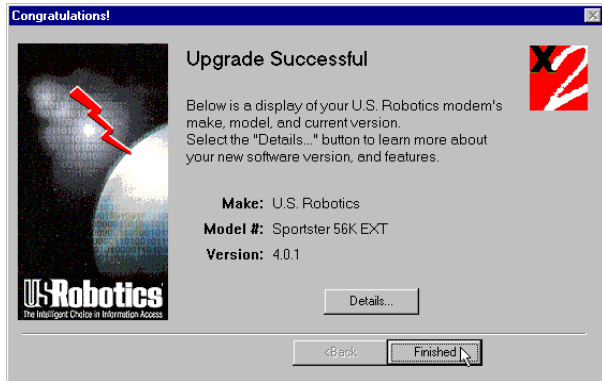
WARNING! To avoid the risk of damaging your modem, DO NOT turn off the modem while it is being updated.

11. When updating is complete, the progress bar will be full and an "Update/Upgrade Complete" message appears. Click **Next**.



U.S. ROBOTICS MODEM UPDATE WIZARD

- 12.** You will see a “Congratulations!” screen.
Click **Details**.



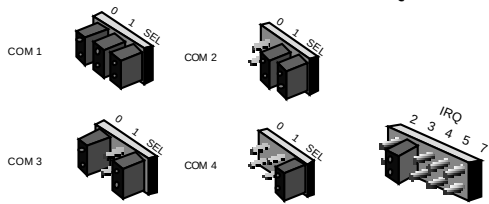
- 13.** Click **OK** once you verify that the process was a success. When you return to the “Congratulations!” screen, click **Finished**.



Congratulations!

Enjoy the benefits of your updated Sportster modem!

TROUBLESHOOTING AND ONLINE HELP RESOURCES

PROBLEM	DIAGNOSIS	POSSIBLE SOLUTION
The computer or software will not recognize the modem.	Your modem might not be turned on.	Make sure the modem is turned on. The power switch is on top of external modems. The CS light on the front panel should be lit. If you installed an internal modem, it should power up when the computer is turned on.
	You may not be entering modem commands in the proper manner.	Type in all upper case (AT) or lower case (at).
	If you are using an external modem, the COM port may not be enabled.	Refer to your computer's manual for information concerning enabling COM ports (usually involves altering the bios settings, motherboard jumpers, and the operating system).
	You may have a COM port/IRQ conflict.	WINDOWS 95 USERS: Uninstall the modem from your computer. In Device Manager, determine what COM port and IRQ settings are free on your system. Set the jumper shunts on your modem's jumper pins to these free settings, using the diagrams below as a guide. If no IRQ setting is available, you may need to move a device off of an IRQ in order to free it for use by your modem. 

TROUBLESHOOTING AND ONLINE HELP RESOURCES

PROBLEM	DIAGNOSIS	POSSIBLE SOLUTION
The computer or software will not recognize the modem (<i>cont.</i>).	You may have a COM port/IRQ conflict (<i>cont.</i>).	Make sure you have the correct COM port and IRQ settings in your software and/or in the Windows Device Manager.
The modem displays double characters on your monitor.	Both the modem's and software's local character echoes are probably turned on.	Disable Local Echo in your software OR on your modem (not both). You can turn the local echo off on the modem by typing ATE0 and then pressing ENTER in your software's terminal mode. To turn the local echo off in the software, refer to its documentation.
The modem won't go off hook to dial or doesn't answer the phone.	You may have plugged your modem's phone cord into a digital line.	Plugging your modem's phone cord into a digital phone line can damage the modem. Call your phone company if you are unsure whether or not your phone line is digital.
	You might have a bad phone cord connection to your modem.	The phone cord should be plugged into the TELCO jack on the modem and the wall phone jack. The phone cord should be no longer than 12 feet in length. Use the phone cord included in your Sportster box if possible.
	You may have devices between the modem and the phone jack.	There should be no line splitters, fax machines, or other devices between the modem and the wall jack.
	You may have a poor line connection.	Place the call again. Calls are routed differently each time. To verify a valid phone connection, enter RapidComm's Terminal mode, type ATX3DT8479855092 (the U.S. Robotics BBS), and press ENTER . This string bypasses the dial tone, allowing a connection if the modem is functioning properly.

TROUBLESHOOTING AND ONLINE HELP RESOURCES

PROBLEM	DIAGNOSIS	POSSIBLE SOLUTION
The modem won't go off hook to dial or doesn't answer the phone (<i>cont.</i>)	If you have voice mail, your dial tone may be altered when messages are waiting.	Answer your voice mail to restore your normal dial tone.
	Your software may not have auto answer enabled.	Enable the auto answer feature. In RapidComm's Terminal mode, type ATS0=1 and press ENTER . NOTE: You need to enable auto answer before every session unless you alter your software's initialization string to permanently enable auto answer.
Both modems exchange carrier signals but fail to establish a link.	You may have a poor line connection.	Try placing the call again. The phone company routes calls differently each time. To verify a valid phone connection, enter RapidComm's Terminal mode and type ATX3DT8479855092 and press ENTER (the phone number is the U.S. Robotics BBS). This string bypasses the dial tone, allowing a connection if the modem is functioning properly. Check http://x2.usr.com/connectnow for a list of ISPs that offer x2 technology.
	Your Internet Service Provider (ISP) may not be x2 capable.	Call your phone company to determine if this might be the problem. You can also run U.S. Robotics' Line Test utility to determine if your phone line is x2 capable. Visit our Line Test Web page at http://x2.usr.com/connectnow/linetest.html .
Your x2 modem cannot achieve an x2 Internet connection.	The phone lines in your area may not be x2 capable.	

TROUBLESHOOTING AND ONLINE HELP RESOURCES

PROBLEM	DIAGNOSIS	POSSIBLE SOLUTION
Your modem won't connect at 2400 bps with a 2400 bps modem.	The modem you're trying to connect with could be an older model that doesn't support error control. You could have a conflict with the remote modem's settings for word length, parity, and stop bits.	You can disable error control on your modem by typing AT&M0 in terminal mode and pressing ENTER. Now try placing the call to the remote modem again. When finished, reset your modem to enable the error control features. In terminal mode, type ATZ and press ENTER. Note: ATZ4 or AT&F1 are often the best reset strings, as they restore hardware flow control defaults. Set your modem's word length, parity, and stop bits the same as the remote modem or BBS you are calling.
Your screen keeps displaying random garbage characters.	Your software and modem might not be set to the same flow control settings. The best flow control settings might not be enabled on your modem.	Make sure the software and modem have the same flow control settings (hardware [RTS/CTS] and software [xon/xoff]). In terminal mode, type AT&F1 and press ENTER to load the optimal settings.

TROUBLESHOOTING AND ONLINE HELP RESOURCES

PROBLEM	DIAGNOSIS	POSSIBLE SOLUTION
Your communications software is reporting many cyclic redundancy check (CRC) errors and low characters per second (CPS).	You may have a bad phone line.	Try placing the call again. The phone company routes calls differently each time.
	Optimum flow control settings may not be enabled on your modem.	In terminal mode, type AT&F1 and press ENTER to load the optimum hardware flow control settings:.
	The serial port rate in your communications software may be set too high for your modem's UART or your area's phone lines.	Lower the serial port rate in your communications software to 57,600 bps, 38,400 bps, or 19,200 bps. NOTE: The port rate for an x2 modem cannot be set lower than 57,600.
	The remote site you are dialing into may have trouble with the file transfer protocol.	Try using a different file transfer protocol. Do not use Xmodem if other protocols are available. Zmodem is the preferred protocol.
	There may be a Terminate and Stay Resident (TSR) program (such as a screen saver or virus scanner) running in the background, disrupting data communications.	Disable any Terminate and Stay Resident (TSR) programs running in the background. If you have software running as a TSR, check the software's manual for information about disabling its ability to operate as a TSR.

TROUBLESHOOTING AND ONLINE HELP RESOURCES

PROBLEM	DIAGNOSIS	POSSIBLE SOLUTION
Your communications software is reporting many cyclic redundancy check (CRC) errors and low characters per second (CPS) (<i>cont.</i>). Errors are constantly occurring in your V.17 fax transmissions.	You may be trying to download a file to a compressed area of your hard disk.	Download to an uncompressed area on your hard drive.
	Your modem initialization string could be insufficient for fax transmissions.	In terminal mode, type AT&H3&I2&R2S7=90 and press ENTER . This is an initialization string. The standard string for faxing is AT&F1S36=0 .
	There may be a Terminate and Stay Resident (TSR) program (such as a screen saver or virus scanner) running in the background, disrupting data communications.	Disable any Terminate and Stay Resident (TSR) programs running in the background. If you have software running as a TSR, check the software's manual for information about disabling its ability to operate as a TSR.
RapidComm fails to initialize the modem.	Your baud rate may be set too high.	Lower baud rate to 9600, 7200, or 4800..
	You may be trying to fax a compressed file	Open the file in the application in which it was created. Select RapidComm as the printer and then print the file.
	RapidComm's port settings may be incorrect.	Make sure RapidComm's port settings are correctly set for your modem.

TROUBLESHOOTING AND ONLINE HELP RESOURCES

Online Help Resources

Connecting to the U.S. Robotics BBS

To connect to the U.S. Robotics Bulletin Board System, follow these steps:

1. Start RapidComm. The software settings for the BBS are:

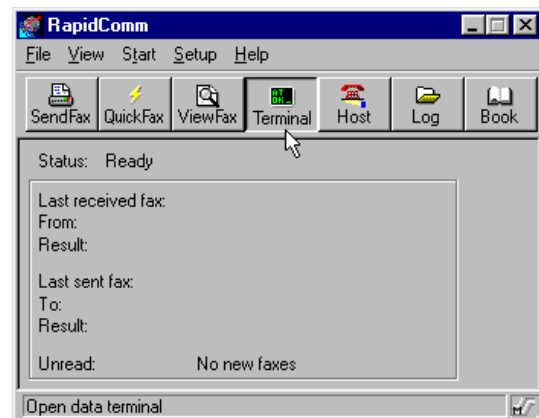
ANSI terminal emulation

Data Bits: 8

Parity: None

Stop Bits: 1

2. Put RapidComm in Terminal mode by clicking on the *Terminal* button (see the following screen image).



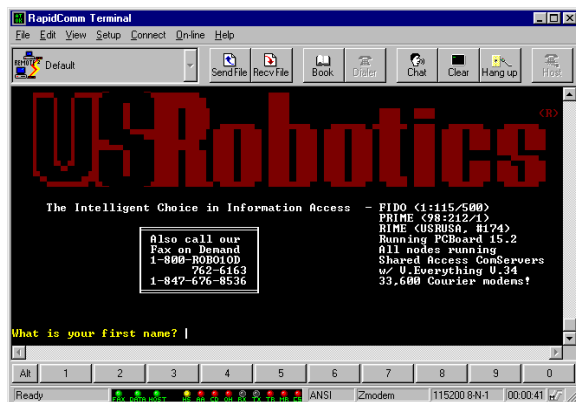
3. Type **ATDS0** and press **ENTER**.

 **NOTE: ATDS0 automatically dials 1-847-982-5092, the U.S. Robotics BBS.**

If this is your first time connecting to our BBS, you will be asked to enter your name,

TROUBLESHOOTING AND ONLINE HELP RESOURCES

create a password of your choice, and to fill out a questionnaire.



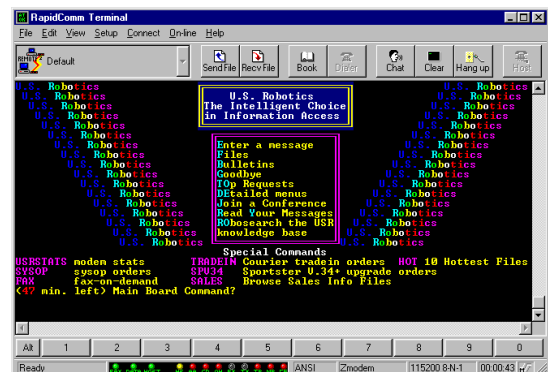
The preceding screen shot is what the introduction screen of the U.S. Robotics BBS will look like. The BBS gives you access to customer and technical support documents and the BBS library. The library

contains hundreds of helpful files and tips to help simplify using your modem.

Downloading the Technical Reference Guide

To download the *Technical Reference Guide*, follow these steps:

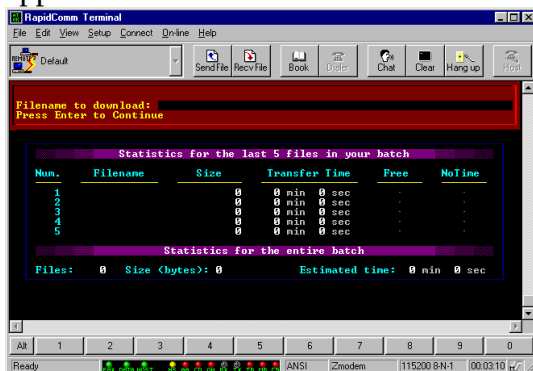
1. In the main menu, type **D** for Download a file.



(This is what the main menu will look like.)

TROUBLESHOOTING AND ONLINE HELP RESOURCES

2. A **Filename to Download:** text box appears on the screen.



3. Type the name of the file you wish to receive. You have three manual formats from which to choose:
 - **x2_WORD.ZIP**—The Guide in a zipped Word for Windows v6.0 format. You will need PKUNZIP.EXE to uncompress this file.

PKUNZIP.EXE is also available on the BBS.

- **x2_HELP.ZIP**—The Guide in a zipped Windows Help format.
 - **x2_ASCII.TXT**—The Guide in an uncompressed ASCII format.
4. **Protocol Type for Transfer.** Your selection depends on what your software supports. If possible, make Zmodem your first choice. Xmodem should be your last choice since it is very slow.
 5. Depending on the software you are using, you will either be prompted where you want the file placed, or the file will be placed in the directory where your communications software is loaded.
 6. When the file transfer is complete, and you are ready to leave the BBS, type **G** (for Good-bye) from the main menu.

TROUBLESHOOTING AND ONLINE HELP RESOURCES

U.S. Robotics offers a number of other online technical support options. Choose any one of the following options if you need help with or want to learn more about your new modem.

Internet FTP

Provides free library containing the same files as the BBS site. FTP to **ftp.usr.com**.

Internet on Demand

Provides automatic technical support through a library containing product information, quick reference cards, and installation help. To obtain an index of available documents, send a blank e-mail to **support@usr.com**. To have a document e-mailed to you, send the document's number as the subject.

World Wide Web

A U.S. Robotics Home Page containing the same information as the Internet on Demand listing. Log on to:

www.usr.com/home/online/main.htm

CompuServe

Access the same information as the Internet FTP site. Connect through the **US Robotics** Forum or e-mail us through CompuServe Mail for a response within 24 hours. The U.S. Robotics forum address is **GO USROBOTICS**. Address private messages to **76711,707**.

America Online

Connect to U.S. Robotics through America Online. Go to the **Keyword** field and type **USROBOTICS** to connect to various U.S. Robotics resources, such as file libraries,

TROUBLESHOOTING AND ONLINE HELP RESOURCES

message boards, online customer support, and product announcements.

Technical Support Hotline

Technical questions about U.S. Robotics modems can also be answered by technical support representatives.

Hotline (847) 982-5151
(Hours: 8:00 am - 6:00 pm CST)

Priority No-Hold Service

U. S. Robotics also staffs its own 900 fee-based number for immediate assistance. These lines are staffed from 8 a.m. - 6 p.m. CST Monday through Friday.

No-Hold line 900-555-USR1

Are You Still Having Problems?

- Review this manual.

- Call or visit your modem dealer. They may be able to assist you. This is much more efficient and time-saving than returning the modem to U.S. Robotics.
- If your dealer can't help you, contact U.S. Robotics Customer Support. When you call, specify your modem serial number (found on the modem and on the outside of the box), the software being used, and, if possible, have the contents of your ATI7 screen available.

If You Need to Return the Modem to Us

Contact U.S. Robotics Customer Support. If the support representative determines that you need to return the modem, you will receive an SRO (Service Repair Order) number. You must have an SRO number

TROUBLESHOOTING AND ONLINE HELP RESOURCES

before returning the modem to us. Ship the unit, postage paid, in a strong box made of corrugated cardboard with plenty of packing material. DO NOT send the modem back in the original box. Send ONLY the modem (NOT the power supply, manuals, CD-ROM, etc.). Include your SRO number, name, and address on the shipping label as well as inside the package. If possible, send the package via a courier capable of tracking the progress of the shipment.

Ship to the following address:

3Com
Attn: RMA
SRO#
6201 W. Oakton, East Dock
Morton Grove, IL 60053

RapidComm Troubleshooting Tips

PROBLEM: *The lights in RapidComm's modem light monitor do not correspond to the actual lights on an external modem.*

DIAGNOSIS: Under certain circumstances, the modem light monitor does not correctly report the activity of the modem. We are aware of this problem and are working to rectify the situation in upcoming versions of the software.

PROBLEM: *The station ID works intermittently.*

DIAGNOSIS: Under certain circumstances (including receiving a fax), the station ID may not work properly.

TROUBLESHOOTING AND ONLINE HELP RESOURCES

PROBLEM: *RapidComm does not work properly at 115.2 kbps.*

DIAGNOSIS: The presence of disk compression software on your system hampers RapidComm's ability to work properly at 115.2 kbps. Removal of the disk compression software from your system will allow RapidComm to operate at 115.2 kbps.

A Note to User's with Older Versions of RapidComm on Their Systems

You must uninstall older versions of RapidComm from your system before installing the most recent version (contained on the CD-ROM that came with your new 56K modem). Otherwise you may introduce conflicts in your system that will hamper RapidComm's ability to work properly.

GLOSSARY

Cross references are printed in **boldface**. Cross references with items in the Command Summary, found in the Technical Quick Reference (Section D), are printed in *italics*.

analog loopback

A modem self-test in which data from the keyboard or an internal test pattern is sent to the modem's transmitter, turned into analog form, looped back to the receiver, and converted back into digital form.

analog signals

A variety of signals and wavelengths that can be transmitted over communications lines such as the sound of a voice over the phone line.

answer mode

The mode used by your modem when answering an incoming call from an originating

modem. The transmit/receive frequencies are the reverse of the originating modem, which is in **originate mode**.

application

A computer program designed to perform a specific task or set of tasks. Examples include word processing and spreadsheet applications.

ARQ

Automatic Repeat reQuest. A function that allows your modem to detect flawed data and request that it be retransmitted. See **MNP** and **V.42**.

ASCII

American Standard Code for Information Interchange. A code used to represent letters, numbers, and special **characters** such as \$, !, and /.

GLOSSARY

asynchronous transmission

Data transmission in which the length of time between transmitted **characters** may vary. Because characters may not be transmitted at set intervals, **start/stop** bits are used to mark the beginning and end of each character.

auto answer

Sets the modem to pick up the phone line when it detects a certain number of rings. See S-register S0 in the “Technical Quick Reference.”

auto dial

A process where your modem dials a call for you. The dialing process is initiated by sending an *ATDT* (dial tone) or *ATDP* (dial pulse) command followed by the telephone number.

Auto dial is used to dial voice numbers. See command *Dn*.

baud rate

A term used to measure the speed of an analog transmission from one point to another. Although not technically accurate, baud rate is commonly used to mean **bit rate**.

binary digit

A 0 or 1, reflecting the use of the binary numbering system. Used because the computer recognizes either of two states, OFF or ON. Shortened form of binary digit is bit.

bit rate

Also referred to as transmission rate. The number of **binary digits**, or bits, transmitted per second (**bps**). Communications channels using analog modems are established at set bit

GLOSSARY

rates, commonly 2400, 4800, 9600, 14,400, 28,800 and higher.

bits per second (bps)

The bits (**binary digits**) per second rate. Thousands of bits per second are expressed as kilobits per second (kbps).

buffer

A temporary memory area used as storage during input and output operations. An example is the modem's command buffer.

byte

A group of **binary digits** stored and operated upon as a unit. Most often the term refers to 8-bit units or **characters**. One kilobyte (KB) is equal to 1,024 bytes or characters; 640 KB is equal to 655,360 bytes or characters.

carrier

The basic signal altered or modulated by the modem in order to carry information.

character

A representation, coded in **binary digits**, of a letter, number, or other symbol.

characters per second (CPS)

A data transfer rate generally estimated from the **bit rate** and the **character** length. For example, at 2400 bps, 8-bit characters with **start/stop bits** (for a total of ten bits per character) will be transmitted at a rate of approximately 240 characters per second (cps). Some **protocols**, such as error-control protocols, employ advanced techniques such as longer transmission **frames** and **data compression** to increase cps.

GLOSSARY

class 1 and 2.0

International standards used by fax **application** programs and faxmodems for sending and receiving faxes.

cyclic redundancy checking (CRC)

An error-detection technique consisting of a test performed on each block or **frame** of data by both sending and receiving modems. The sending modem inserts the results of its tests in each data block in the form of a CRC code. The receiving modem compares its results with the received CRC code and responds with either a positive or negative acknowledgment.

data communications

The transmission or sharing of data between computers via an electronic medium.

data compression table

A table containing values assigned for each **character** during a call under **MNP5** data compression. **Default** values in the table are continually altered and built during each call: The longer the table, the more efficient **throughput** gained.

data mode

Mode used by a modem when sending and receiving data files.

DCE

Data Communications (or Circuit-Terminating) Equipment, such as dial-up modems that establish and control the data link via the telephone network.

GLOSSARY

default

Any setting assumed, at startup or reset, by the computer's software and attached devices. The computer or software will use these settings until changed by the user or other software.

detect phase

In the **ITU-T V.42** error-control **protocol**, the first stage in establishing if both modems attempting to connect have **V.42** capability.

dictionary

The term used for compression codes built by the **V.42 bis** data compression algorithm.

digital loopback

A test that checks the modem's RS-232 interface and the cable that connects the **terminal** or computer and the modem. The modem receives data (in the form of **digital**

signals) from the computer or terminal, and immediately returns the data to the screen for verification.

digital signals

Discrete, uniform signals. In this manual, the term refers to the **binary digits** 0 and 1.

DTE

Data **Terminal** (or Terminating) Equipment. A computer that generates or is the final destination of data.

duplex

Indicates a communications channel capable of carrying signals in both directions. See **half duplex**, **full duplex**.

GLOSSARY

Electronic Industries Association (EIA)

Group which defines electronic standards in the U.S.

error control

Various techniques that check the reliability of **characters (parity)** or blocks of data. **V.42** and **MNP** error-control **protocols** use error detection (**CRC**) and retransmission of flawed **frames (ARQ)**.

facsimile

A method for transmitting the image on a page from one point to another. Commonly referred to as fax.

fax mode

The mode used by a modem to send and receive data in **facsimile** format. See definitions for **V.17**, **V.27ter**, **V.29**.

flow control

A mechanism that compensates for differences in the flow of data into and out of a modem or other device. See commands **&Hn**, **&In**, **&Rn**.

frame

A **data communications** term for a block of data with header and trailer information attached. The added information usually includes a frame number, block size data, error-check codes, and Start/End indicators.

full duplex

Signals can flow in both directions at the same time over one line. In microcomputer communications, may refer to the suppression of the online **local echo**.

GLOSSARY

half duplex

Signals can flow in both directions, but only one way at a time. In microcomputer communications, may refer to activation of the online **local echo**, which causes the modem to send a copy of the transmitted data to the screen of the sending computer.

Hz

Hertz, a frequency measurement unit used internationally to indicate cycles per second.

ITU-T

An international organization that defines standards for telegraphic and telephone equipment. For example, the Bell 212A standard for 1200-bps communication in North America is observed internationally as ITU-T **V.22**. For 2400-bps communication, most U.S. manufacturers observe V.22 bis.

LAPM

Link Access Procedure for Modems. An error-control **protocol** defined in **ITU-T** Recommendation V.42. Like the **MNP** protocols, LAPM uses **cyclic redundancy checking (CRC)** and retransmission of corrupted data (**ARQ**) to ensure data reliability.

local echo

A modem feature that enables the modem to display keyboard commands and transmitted data on the screen. See command *En*.

GLOSSARY

MNP

Microcom Networking Protocol, an error-control **protocol** developed by Microcom, Inc., and now in the public domain. There are several different MNP protocols, but the most commonly used one ensures error-free transmission through error detection (**CRC**) and retransmission of flawed **frames**.

modem

A device that transmits/receives computer data through a communications channel such as radio or telephone lines. It also changes signals received from the phone line back to **digital signals** before passing them to the receiving computer.

nonvolatile memory (NVRAM)

User-programmable random access memory whose data is retained when power is turned

off. On the Sportster modem, it includes four stored phone numbers and the modem settings.

off/on hook

Modem operations that are the equivalent of manually lifting a phone receiver (taking it off-hook) and replacing it (going on-hook).

online fall back/fall forward

A feature that allows high-speed, error-control modems to monitor line quality and fall back to the next lower speed in a defined range if line quality diminishes. As line conditions improve, the modems switch up to the next higher speed.

originate mode

The mode used by your modem when initiating an outgoing call to a destination modem. The transmit/receive frequencies are the reverse of the called modem, which is in **answer mode**.

GLOSSARY

parity

A simple error-detection method that checks the validity of a transmitted **character**. Character checking has been surpassed by more reliable and efficient forms of error checking, including **V.42** and **MNP 2-4 protocols**. Either the same type of **parity** must be used by two communicating computers, or both may omit parity.

protocol

A system of rules and procedures governing communications between two or more devices. Protocols vary, but communicating devices must follow the same protocol in order to exchange data. The format of the data, readiness to receive or send, error detection and error correction are some of the operations that may be defined in protocols.

RAM

Random Access Memory. Memory that is available for use when the modem is turned on, but that clears of all information when the power is turned off. The modem's RAM holds the current operational settings, a **flow control buffer**, and a command **buffer**.

remote digital loopback

A test that checks the phone link and a remote modem's transmitter and receiver.

remote echo

A copy of the data received by the remote system, returned to the sending system, and displayed on the screen. Remote echoing is a function of the remote system.

GLOSSARY

ROM

Read Only Memory. Permanent memory, not user-programmable.

serial transmission

The consecutive flow of data in a single channel. Compare to parallel transmissions where data flows simultaneously in multiple channels.

start/stop bits

The signaling bits attached to a **character** before and after the character is transmitted during **asynchronous transmission**.

terminal

A device whose keyboard and display are used for sending and receiving data over a communications link. Differs from a

microcomputer or a mainframe in that it has little or no internal processing capabilities.

terminal mode

Software mode that allows direct communication with the modem. Also known as command mode.

throughput

The amount of actual user data transmitted per second without the overhead of **protocol** information such as **start/stop bits** or **frame** headers and trailers. Compare with **characters per second**.

V.8

The **ITU-T** standard specification that covers the initial handshaking process.

GLOSSARY

V.17 fax

An **ITU-T** standard for making **facsimile** connections at 14,400 bps, 12,000 bps, 9,600 bps, 7,200 bps.

V.21

An **ITU-T** standard for modems operating in asynchronous mode at speeds up to 300 bps, **full-duplex**, on public switched telephone networks.

V.22

An **ITU-T** standard for modem communications at 1200 bps, compatible with the Bell 212A standard observed in the U.S. and Canada.

V.22 bis

An **ITU-T** standard for modem communications at 2400 bps. The standard

includes an automatic link negotiation fallback to 1200 bps and compatibility with Bell 212A/V.22 modems.

V.27 ter

An **ITU-T** standard for **facsimile** operations that specifies modulation at 4800 bps, with fallback to 2400 bps.

V.29

An **ITU-T** standard for **facsimile** operations that specifies modulation at 9600 bps, with fallback to 7200 bps.

V.32

An **ITU-T** standard for modem communications at 9600 bps and 4800 bps. V.32 modems fall back to 4800 bps when line quality is impaired.

GLOSSARY

V.32 bis

An **ITU-T** standard that extends the V.32 connection range: 4800, 7200, 9600, 12,000, and 14,400 bps. V.32 *bis* modems fall back to the next lower speed when line quality is impaired, fall back further as necessary, and also fall forward (switch back up) when line conditions improve (see **online fall back/fall forward**).

V.34

An **ITU-T** standard that currently allows data rates as high as 28,800 bps.

V.34+

An enhancement to **V.34** that enables data transfer rates as high as 33,600 bps.

V.42

An **ITU-T** standard for modem communications that defines a two-stage process of detection and negotiation for **LAPM error control**.

V.42 bis

An extension of **ITU-T** V.42 that defines a specific data compression scheme for use during V.42 connections.

Xmodem

The first of a family of **error control** software **protocols** used to transfer files between modems. These protocols are in the public domain and are available from many bulletin board services.

GLOSSARY

XON/XOFF

Standard **ASCII** control **characters** used to tell an intelligent device to stop/resume transmitting data.

Ymodem

An error-checking **protocol** that can send several files of data at a time in 1024-**byte** (1K) blocks. This protocol can use either checksums or CRC for error checking.

Ymodem G

Similar to **Ymodem**, except it includes no error checking, which makes it faster.

Zmodem

Similar to **Xmodem** and **Ymodem**, except it includes batch transfer, the ability to recover from a partially complete transfer, an autostart feature, and improved efficiency.

TECHNICAL QUICK REFERENCE

This section includes information about:

- Σ *Front Panel Lights*
- Σ *Command Summary*
- Σ *DIP Switches*
- Σ *S-Registers*
- Σ *The Serial Interface (cable information)*

TECHNICAL QUICK REFERENCE

Front Panel Lights (External Modems)

Symbol	Meaning	Status
AA	Auto Answer.....	Answer mode: ON when register S0 is set to 1 or higher (Auto Answer), and when answering a call; OFF when modem originates a call. Light flashes when there is an incoming call.
CD.....	Carrier Detect	ON if modem receives a valid data signal (carrier) from a remote modem, indicating that data transmission is possible. Always ON if CD override is ON (&C0).
RD.....	Received Data.....	Flashes when modem sends result codes or passes received data bits from remote.
SD	Send Data	Flashes when computer sends a data bit to modem.
TR.....	Data Terminal Ready	ON if modem receives a DTR signal from computer. Always ON (modem ignores DTR) if the DTR override is ON (&D0).
CS	Clear to Send.....	ON until modem lowers CTS when Transmit Data hardware flow control is enabled (&H1, &H3).
ARQ/	Error Control/	
FAX.....	Fax Operations	Data Mode: Automatic Repeat Request. ON if modem is set to &M4 or &M5 and successfully establishes an error control connection. Flashes when modem retransmits data to remote modem. Fax Mode: Flashes to indicate fax mode.

TECHNICAL QUICK REFERENCE

Command Summary

- Type commands in either upper or lower case, not a combination. Use the Backspace key to delete errors. (You cannot delete the original AT command since it is stored in the modem buffer.)
- If a command has numeric options and you don't include a number, zero is assumed. For example, if you type **ATB**, the command **ATB0** is assumed.
- Every command except A/ and +++ must begin with the AT prefix and be entered by pressing **ENTER**.
- The maximum command length is 58 characters. The modem doesn't count the AT prefix, carriage returns, or spaces.



NOTE: All defaults are based on the &F1—Hardware Flow Control template loaded in NVRAM when the modem is shipped. Defaults are listed in *italics*.

TECHNICAL QUICK REFERENCE

Command Set

- \$** Use in conjunction with *D*, *S*, or *&* commands (or just *AT*) to display a basic command list; online help.
- A** Manual Answer: goes off hook in answer mode. Pressing any key aborts the operations.
- A/** Re-executes the last issued command. Used mainly to redial. This does not require the *AT* prefix or a Carriage Return.
- Any key** Aborts off-hook dial/answer operation and hangs up.
- AT** Required command prefix, except with *A/* and *+++*. Use alone to test for OK result code.

Bn U.S./ITU-T answer sequence.

B0 ITU-T answer sequence

B1 U.S. answer tone

Dn Dials the specified phone number. Includes the following:

L Dials the last dialed number.

P Pulse (rotary) dial

R Originates call using answer (reverse) frequencies.

Sn Dials the phone number string stored in NVRAM at position *n* (*n* = 0–3). Phone

TECHNICAL QUICK REFERENCE

	numbers are stored with the &Zn=s command.		
T	Tone dial	W	Wait for second dial tone (X2 or X4); linked to S6 register.
,	(Comma) Pause, See S8 definition; which it's linked to.	@	Dials, waits for quiet answer, and continues (X3 or higher).
;	(Semicolon) Return to Command mode after dialing.	\$	Displays a list of Dial commands.
“	Dials the letters that follow (in an alphabetical phone number).	En	Sets local echo.
!	(Exclamation point) Flashes the switch hook.	E0	Echo OFF
/	Delays for 125 ms. before proceeding with dial string.	E1	<i>Modem displays keyboard commands</i>
		Fn	Sets online local echo of transmitted data ON/OFF.
		F0	Local echo ON. Modem sends a copy of data it sends to the

TECHNICAL QUICK REFERENCE

	remote system to your screen.	I2	Results of RAM checksum
F1	<i>Local echo OFF. Receiving system may send a remote echo of data it receives.</i>	I3	Product type
		I4	Current modem settings
		I5	Nonvolatile memory (NVRAM) settings
Hn	Controls ON/OFF hook.	I6	Link diagnostics
H0	Hangs up (goes on hook).	I7	Product configuration
H1	Goes off hook.	Ln	Controls speaker volume (internals only).
In	Displays the following information.	L0	Low volume
I0	Four-digit product code	L1	Low volume
I1	Results of ROM checksum	L2	<i>Medium volume</i>
		L3	High volume
		Mn	Operates speaker.
		M0	Speaker always OFF.

TECHNICAL QUICK REFERENCE

<i>M1</i>	<i>Speaker ON until CONNECT.</i>	<i>Q2</i>	Displays result codes only in Originate mode.
<i>M2</i>	Speaker always ON.	<i>Sr.b=n</i>	Sets bit .b of register r to n (0/OFF or 1/ON).
<i>M3</i>	Speaker ON after dial, until CONNECT.	<i>Sr=n</i>	Sets register r to n.
On	Returns online.	<i>Sr?</i>	Displays contents of S-Register r.
<i>O0</i>	Returns online.	<i>S\$</i>	Displays a list of the S-Registers.
<i>O1</i>	Returns online and retrains.	<i>T</i>	Sets tone dial.
P	Sets pulse dial (for phone lines that don't support touch-tone dialing).	<i>Vn</i>	Displays verbal/numeric result codes.
Qn	Displays/suppresses result codes.	<i>V0</i>	Numeric codes
<i>Q0</i>	<i>Displays result codes.</i>	<i>V1</i>	<i>Verbal codes</i>
<i>Q1</i>	Quiet mode; no result codes.		

TECHNICAL QUICK REFERENCE

Xn **Sets result code displayed. Default is X4.**

Result Codes	Xn Setting				
	X0	X1	X2	X3	X4
0/OK	•	•	•	•	•
1/CONNECT	•	•	•	•	•
2/RING	•	•	•	•	•
3/NO CARRIER	•	•	•	•	•
4/ERROR	•	•	•	•	•
5/CONNECT 1200		•	•	•	•
6/NO DIAL TONE			•		•
7/BUSY				•	•
8/NO ANSWER*				•	•
9/Reserved					
10/CONNECT 2400		•	•	•	•
11/RINGING					•
13/CONNECT 9600		•	•	•	•
18/CONNECT 4800		•	•	•	•
20/CONNECT 7200		•	•	•	•
21/CONNECT 12000		•	•	•	•
25/CONNECT 14400		•	•	•	•
43/CONNECT 16800		•	•	•	•
85/CONNECT 19200		•	•	•	•
91/CONNECT 21600		•	•	•	•
99/CONNECT 24000		•	•	•	•
103/CONNECT 26400		•	•	•	•
107/CONNECT 28800		•	•	•	•
151/CONNECT 31200		•	•	•	•

TECHNICAL QUICK REFERENCE

Xn (continued)

<u>Xn</u>	<u>Result Codes (cont.)</u>	<u>X0</u>	<u>X1</u>	<u>X2</u>	<u>X3</u>	<u>X4</u>
	155/CONNECT 33600	•	•	•	•	•
	180/CONNECT 33333	•	•	•	•	•
	184/CONNECT 37333	•	•	•	•	•
	188/CONNECT 41333	•	•	•	•	•
	192/CONNECT 42666	•	•	•	•	•
	196/CONNECT 44000	•	•	•	•	•
	200/CONNECT 45333	•	•	•	•	•
	204/CONNECT 46666	•	•	•	•	•
	208/CONNECT 48000	•	•	•	•	•
	212/CONNECT 49333	•	•	•	•	•
	216/CONNECT 50666	•	•	•	•	•
	220/CONNECT 52000	•	•	•	•	•
	224/CONNECT 53333	•	•	•	•	•
	228/CONNECT 54666	•	•	•	•	•
	232/CONNECT 56000	•	•	•	•	•
	236/CONNECT 57333	•	•	•	•	•
	Adaptive Dialing			•	•	•
	Wait for 2nd Dial Tone (W)			•		•
	Wait for Answer (@)				•	•
	Fast Dial			•		•
	*Requires @ in dial string; replaces NO CARRIER					

TECHNICAL QUICK REFERENCE

Yn Selects power-on/reset default configuration.

- Y0 Default is profile 0 setting in NVRAM
- Y1 Default is profile 1 setting in NVRAM

Z Resets modem.

- Z0 Resets modem to NVRAM profile selected by Y command or dip 7.
- Z1 Resets modem to NVRAM profile 0
- Z2 Resets modem to NVRAM profile 1
- Z3 Resets modem to factory default profile 0 (&F0)

- Z4 Resets modem to factory default profile 1 (&F1)
- Z5 Resets modem to factory default profile 2 (&F2)

&\$ Displays a list of ampersand (&) commands.

&A Enables/disables additional result code subsets (see Xn).

- &A0 ARQ result codes disabled
- &A1 ARQ result codes enabled
- &A2 V.32 modulation indicator added

TECHNICAL QUICK REFERENCE

&A3	<i>Protocol indicators addedæ LAPM/MNP/NONE (error control) and V42bis/MNP5 (data compression)</i>	&C1	<i>Normal CD operations</i>
&Bn	Manages modem's serial port rate.	&Dn	Controls Data Terminal Ready (DTR) operations.
&B0	<i>Variable, follows connection rate</i>	&D0	<i>DTR override</i>
&B1	<i>Fixed serial port rate</i>	&D1	<i>DTR toggle causes online Command mode</i>
&B2	<i>Fixed in ARQ mode, variable in non-ARQ mode</i>	&D2	<i>Normal DTR operations</i>
&Cn	Controls Carrier Detect (CD) signal.	&D3	<i>Resets on receipt of DTR</i>
&C0	<i>CD override</i>	&Fn	Loads a read-only (non-programmable) factory configuration.
		&F0	<i>Generic template</i>
		&F1	<i>Hardware flow control template</i>

TECHNICAL QUICK REFERENCE

&F2 Software flow control
template

&Gn Sets Guard Tone.

&G0 *No guard tone, U.S.
and Canada*

&G1 550 Hz guard tone,
some European
countries, requires B0
setting.

&G2 1800 Hz guard tone,
U.K., requires B0
setting.

**&Hn Sets Transmit Data (TD)
flow control (see also &Rn).**

&H0 Flow control disabled

&H1 *Hardware flow
control, Clear to
Send (CTS)*

&H2 Software flow control,
XON/XOFF

&H3 Hardware and
software flow control

**&In Sets Receive Data (RD)
software flow control (see
also &Rn).**

&I0 *Software flow
control disabled*

&I1 XON/XOFF signals to
your modem and
remote system

&I2 XON/XOFF signals to
your modem only

**&Kn Enables/disables data
compression.**

&K0 Data compression
disabled

TECHNICAL QUICK REFERENCE

<i>&K1</i>	<i>Auto enable/disable</i>	Sets ceiling connect speed if &Un is greater than 0. See &Un.
<i>&K2</i>	Data compression enabled	
<i>&K3</i>	MNP5 compression disabled	<i>&N0</i> Variable rate
&Mn	Sets Error Control (ARQ) for connections at 1200 bps and higher.	<i>&N1</i> 300 bps
<i>&M0</i>	Normal mode, error control disabled	<i>&N2</i> 1200 bps
<i>&M1</i>	Reserved	<i>&N3</i> 2400 bps
<i>&M2</i>	Reserved	<i>&N4</i> 4800 bps
<i>&M3</i>	Reserved	<i>&N5</i> 7200 bps
<i>&M4</i>	Normal/ARQ	<i>&N6</i> 9600 bps
<i>&M5</i>	ARQ mode	<i>&N7</i> 12,000 bps
&Nn	Sets connect speed. If connection cannot be established at this speed, the modem will hang up.	<i>&N8</i> 14,400 bps
		<i>&N9</i> 16,800 bps
		<i>&N10</i> 19,200 bps
		<i>&N11</i> 21,600 bps
		<i>&N12</i> 24,000 bps
		<i>&N13</i> 26,400 bps
		<i>&N14</i> 28,800 bps
		<i>&N15</i> 31,200 bps

TECHNICAL QUICK REFERENCE

&N16 33,600 bps
&N17 33,333 bps
&N18 37,333 bps
&N19 41,333 bps
&N20 42,666 bps

&N Continued

&N21 44,000 bps
&N22 45,333 bps
&N23 46,666 bps
&N24 48,000 bps
&N25 49,333 bps
&N26 50,666 bps
&N27 52,000 bps
&N28 53,333 bps
&N29 54,666 bps
&N30 56,000 bps
&N31 57,333 bps

**&Pn Sets pulse (rotary) dial
make/break ratio.**

*&P0 U.S./Canada ratio,
39%/61%*

&P1 U.K. ratio, 33%/67%

**&Rn Sets Receive Data (RD)
hardware flow control,
Request to Send (RTS) (see
also &Hn).**

&R0 Reserved

&R1 Modem ignores RTS

*&R2 Received Data to
computer only on
RTS*

TECHNICAL QUICK REFERENCE

&Sn Controls Data Set Ready (DSR) operations.

- &S0** *DSR override; always ON*
- &S1** Modem controls DSR

&Tn Begins test modes.

- &T0** Ends testing
- &T1** Analog Loopback
- &T2** Reserved
- &T3** Local Digital Loopback
- &T4** Enables Remote Digital Loopback
- &T5** *Prohibits Remote Digital Loopback*
- &T6** Initiates Remote Digital Loopback

- &T7** Remote Digital with self-test and error detector

- &T8** Analog Loopback with self-test and error detector

&Un Sets floor connect speed when &Un is set greater than 0. &Nn is the ceiling connect speed. See &Nn.

- &U0** *Disabled*
- &U1** 300 bps
- &U2** 1200 bps
- &U3** 2400 bps
- &U4** 4800 bps
- &U5** 7200 bps
- &U6** 9600 bps
- &U7** 12,000 bps
- &U8** 14,400 bps

TECHNICAL QUICK REFERENCE

&U9 16,800 bps
&U10 19,200 bps
&U11 21,600 bps
&U12 24,000 bps
&U13 26,400 bps
&U14 28,800 bps
&U15 31,200 bps
&U16 33,600 bps
&U17 33,333 bps
&U18 37,333 bps
&U19 41,333 bps
&U20 42,666 bps
&U21 44,000 bps
&U22 45,333 bps

&U Continued

&U23 46,666 bps
&U24 48,000 bps
&U25 49,333 bps
&U26 50,666 bps
&U27 52,000 bps

&U28 53,333 bps
&U29 54,666 bps
&U30 56,000 bps
&U31 57,333 bps

&Wn Writes current configuration to NVRAM templates.

&W0 Modifies the NVRAM
0 template (Y0)
&W1 Modifies the NVRAM
1 template (Y1)

TECHNICAL QUICK REFERENCE

&Yn Sets break handling.

&Y0 Destructive, but
doesn't send break

&Y1 *Destructive,*
expedited

&Y2 Nondestructive,
expedited

&Y3 Nondestructive,
unexpedited

&Zn=s Writes phone number string
s to NVRAM at position *n*
(*n* = 0–3).

&Zn=L Writes last executed dial
string to NVRAM at
position *n* (*n* = 0–3).

&Zn? Displays the phone number
stored at position *n* (*n* = 0–
3).

&ZL? Displays the last executed
dial string.

#CID=*n* Controls Caller ID feature.

#CID=0 *Caller ID disabled.*

#CID=1 Caller ID enabled.

#CID=2 Caller ID enabled with
extended information,
including caller's name.

+++ Escapes to online-command
mode.

TECHNICAL QUICK REFERENCE

DIP Switches (Modems with DIP Switches Only)

Note: If a DIP switch is on, it is down. If a DIP switch is off, it is up. *Defaults are in italics.*

Switch	Factory Setting	Function
1	OFF	Data Terminal Ready (DTR) Override OFF Normal DTR operations: computer must provide DTR signal for the modem to accept commands; dropping DTR terminates a call ON Modem ignores DTR (Override)
2	OFF	Verbal/Numeric Result Codes OFF Verbal (word) results ON Numeric results
3	ON	Result Code Display OFF Suppresses result codes ON Enables result codes
4	OFF	Command Mode Local Echo Suppression OFF Displays keyboard commands ON Suppresses echo
5	ON	Auto Answer Suppression OFF Modem answers on first ring, or higher if specified in NVRAM

TECHNICAL QUICK REFERENCE

		ON	Disables auto answer
Switch	Factory Setting	Function	
6	OFF		Carrier Detect (CD) Override
		OFF	Modem sends CD signal when it connects with another modem, drops CD on disconnect
		ON	CD always ON (Override)
7	OFF		Power-on and ATZ Reset Software Defaults
		OFF	Loads Y or Y1 configuration from user-defined nonvolatile memory (NVRAM)
8	ON	ON	Loads &F0—Generic template from read only memory (ROM)
			AT Command Set Recognition
		OFF	Disables command recognition (Dumb Mode)
		ON	Enables recognition (Smart mode)

TECHNICAL QUICK REFERENCE

S-Registers

To change a setting, use the $ATSr=n$ command, where r is the register and n is a decimal value from 0 – 255 (unless otherwise indicated).

Register	Default	Function
S0	0	Sets the number of rings on which to answer in Auto Answer Mode. When set to 0, Auto Answer is disabled.
S1	0	Counts and stores the number of rings from an incoming call. (S0 must be greater than 0.)
S2	43	Stores the ASCII decimal code for the escape code character. Default character is +. A value of 128 – 255 disables the escape code.
S3	13	Stores the ASCII code for the Carriage Return character. Valid range is 0 – 127.
S4	10	Stores the ASCII decimal code for the Line Feed character. Valid range is 0 – 127.
S5	8	Stores the ASCII decimal code for the Backspace character. A value of 128 – 255 disables the Backspace key's delete function.

TECHNICAL QUICK REFERENCE

Register	Default	Function
S6	2	Sets the number of seconds the modem waits before dialing. If Xn is set to X2 or X4, this is the time-out length if there isn't a dial tone.
S7	60	Sets the number of seconds the modem waits for a carrier. May be set for much longer duration if, for example, the modem is originating an international connection.
S8	2	Sets the duration, in seconds, for the pause (,) option in the Dial command.
S9	6	Sets the required duration, in tenths of a second, of the remote modem's carrier signal before recognition by the Sportster modem.
S10	7	Sets the duration, in tenths of a second, that the modem waits to hang up after loss of carrier. This guard time allows the modem to distinguish between a line disturbance from a true disconnect (hang up) by the remote modem. While we don't recommend connecting the modem to a line with call waiting, if you have it, you may wish to adjust this setting upward to prevent the modem from misinterpreting the second call signal as a disconnect by the remote modem.

TECHNICAL QUICK REFERENCE

Register	Default	Function															
S10 (<i>cont.</i>)		A better alternative is to ask your phone company how to temporarily disable call waiting (usually *70W). For example: ATDT *70W <i>phone number</i>. Note: If you set S10 = 255, the modem will not hang up when carrier is lost. Dropping DTR hangs up the modem.															
S11	70	Sets the duration and spacing, in milliseconds, for tone dialing.															
S12	50	Sets the duration, in fiftieths of a second, of the guard time for the escape code sequence (+++).															
S13	0	Bit-mapped register. Select the bit(s) you want on and set S13 to the total of the values in the Value column. For example, ATS13 = 17 enables bit 0 (value is 1) and bit 4 (value is 16). <table> <tr> <th>Bit</th><th>Value</th><th>Result</th></tr> <tr> <td>0</td><td>1</td><td>Reset when DTR drops.</td></tr> <tr> <td>1</td><td>2</td><td>Reset non-MNP transmit buffer from 1.5K to 128 bytes.*</td></tr> <tr> <td>2</td><td>4</td><td>Set backspace key to delete.</td></tr> <tr> <td>3</td><td>8</td><td>On DTR signal, auto dial the number stored in NVRAM at position 0.</td></tr> </table>	Bit	Value	Result	0	1	Reset when DTR drops.	1	2	Reset non-MNP transmit buffer from 1.5K to 128 bytes.*	2	4	Set backspace key to delete.	3	8	On DTR signal, auto dial the number stored in NVRAM at position 0.
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3	8	On DTR signal, auto dial the number stored in NVRAM at position 0.															

TECHNICAL QUICK REFERENCE

Register	Default	Function		
S13 (cont.)		Bit	Value	Result
		4	16	At power on/reset, Auto Dial the number stored in NVRAM at position 0.
		5	32	Reserved
		6	64	Disable quick retrains.
		7	128	Disconnect on escape code.
* The 1.5K-byte non-ARQ buffer allows data transfer with Xmodem- and Ymodem-type file protocols without using flow control. The 128-byte option lets remote users with slower modems keep data you're sending from off their screens. When remote users send your computer an XOFF (Ctrl-S) and you stop the data in transit from your modem's buffer doesn't exceed the size of their screen. This is also very helpful in situations when a remote modem/printer application is losing ch				
S14	0	Reserved		
S15	0	Bit-mapped register setup. To set the register, see instructions for S13.		
		Bit	Value	Result
		0	1	Disable ARQ/MNP for V.22.
		1	2	Disable ARQ/MNP for V.22bis.
		2	4	Disable ARQ/MNP V.32/V.32bis/V.32terbo.

TECHNICAL QUICK REFERENCE

Register	Default	Function
S15 (cont.)		Bit Value Result
		3 8 Disable MNP handshake.
		4 16 Disable MNP level 4.
		5 32 Disable MNP level 3.
		6 64 MNP incompatibility.
		7 128 Disable V.42 operation.
		To disable V.42 detect phase, select the total of the values for bits 3 and 7.
S16	0	Bit-mapped register setup. To set the register, see instructions for S13.
		Bit Value Result
		0 1 Reserved
		1 2 Touch tone dialing test.
		2-7 4-128 Reserved
S17	0	Reserved
S18	0	Test timer for &T loopback testing. Sets the time in seconds of testing before the modem automatically times out and terminates the test. When set to 0, the timer is disabled. Valid range is 1-255.

TECHNICAL QUICK REFERENCE

Register	Default	Function
S19	0	Sets the duration, in minutes, for the inactivity timer. The timer activates when there is no data activity on the phone line; at time-out the modem hangs up. S19 = 0 disables the timer.
S20	0	Reserved
S21	10	Sets the length, in 10-millisecond units, of breaks sent from the modem to the computer; applies to MNP or V.42 mode only.
S22	17	Stores the ASCII decimal code for the XON character.
S23	19	Stores the ASCII decimal code for the XOFF character.
S24	0	Reserved
S25	20	Sets the duration, in hundredths of a second, that DTR must be dropped so that the modem doesn't interpret a random glitch as a DTR loss. (Most users will want to use the default; this register is useful for setting compatibility with older systems running under older operating software.)
S26	0	Reserved

TECHNICAL QUICK REFERENCE

Register	Default	Function		
S27	0	Bit-mapped register setup. To set the register, see instructions for S13.		
		Bit	Value	Result
		0	1	Enables ITU-T V.21 modulation at 300 bps for overseas calls; in V.21 mode, the modem answers both overseas and domestic (U.S. and Canada) calls, but only originates V.21 calls. (Default Bell 103)
		1	2	Enables unencoded (non-trellis coded) modulation in V.32 mode.
		2	4	Disables V.32 modulation.
		3	8	Disables 2100 Hz answer tone to allow two V.42 modems to connect faster.
		4	16	Enables V.23 fallback mode.
		5	32	Disables V.32 <i>bis</i> mode.
		6	64	Disable V.42 selective reject.
		7	128	Software compatibility mode. This setting disables the codes and displays the 9600 code instead. The actual rate of the call can be viewed on the ATI6 screen. Used for unusual software incompatibilities. Some software may not accept 7200, 12,000, and 14,400 bps or greater result codes.

TECHNICAL QUICK REFERENCE

Register	Default	Function																								
S28	0	Eliminates the V.32 answer tones for a faster connection.																								
	8	Default item, all times are in tenths of seconds.																								
	255	Disables all connections except V.32 at 9600 bps.																								
S29	20	Sets the duration, in tenths of a second, of the V.21 answer mode fallback timer.																								
S30	0	Reserved																								
S31	128	Reserved																								
S32	2	Bit-mapped register setup. To set the register, see the instructions for S13.																								
<table> <tr> <th>Bit</th><th>Value</th><th>Result</th></tr> <tr> <td>0</td><td>1</td><td>V.8 Call Indicate enabled.</td></tr> <tr> <td>1</td><td>2</td><td>Enables V.8 mode.</td></tr> <tr> <td>2</td><td>4</td><td>Reserved.</td></tr> <tr> <td>3</td><td>8</td><td>Disable V.34 modulation.</td></tr> <tr> <td>4</td><td>16</td><td>Disable V.34+ modulation.</td></tr> <tr> <td>5</td><td>32</td><td>Disable x2 modulation.</td></tr> <tr> <td>6-7-</td><td>64-128</td><td>Reserved</td></tr> </table>			Bit	Value	Result	0	1	V.8 Call Indicate enabled.	1	2	Enables V.8 mode.	2	4	Reserved.	3	8	Disable V.34 modulation.	4	16	Disable V.34+ modulation.	5	32	Disable x2 modulation.	6-7-	64-128	Reserved
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5	32	Disable x2 modulation.																								
6-7-	64-128	Reserved																								

TECHNICAL QUICK REFERENCE

Register	Default	Function																											
S33	0	Bit-mapped register setup. To set the register, see the instructions for S13.																											
		<table> <tr> <th>Bit</th><th>Value</th><th>Result</th></tr> <tr> <td>0</td><td>1</td><td>Disable 2400 symbol rate.</td></tr> <tr> <td>1</td><td>2</td><td>Disable 2743 symbol rate.</td></tr> <tr> <td>2</td><td>4</td><td>Disable 2800 symbol rate.</td></tr> <tr> <td>3</td><td>8</td><td>Disable 3000 symbol rate.</td></tr> <tr> <td>4</td><td>16</td><td>Disable 3200 symbol rate.</td></tr> <tr> <td>5</td><td>32</td><td>Disable 3429 symbol rate.</td></tr> <tr> <td>6</td><td>64</td><td>Reserved</td></tr> <tr> <td>7</td><td>128</td><td>Disable shaping.</td></tr> </table>	Bit	Value	Result	0	1	Disable 2400 symbol rate.	1	2	Disable 2743 symbol rate.	2	4	Disable 2800 symbol rate.	3	8	Disable 3000 symbol rate.	4	16	Disable 3200 symbol rate.	5	32	Disable 3429 symbol rate.	6	64	Reserved	7	128	Disable shaping.
Bit	Value	Result																											
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S34	0	Bit-mapped register setup. To set registers, see instructions for S13.																											
		<table> <tr> <th>Bit</th><th>Value</th><th>Result</th></tr> <tr> <td>0</td><td>1</td><td>Disable 8S-2D trellis encoding.</td></tr> <tr> <td>1</td><td>2</td><td>Disable 16S-4D trellis encoding.</td></tr> <tr> <td>2</td><td>4</td><td>Disable 32S-2D trellis encoding.</td></tr> <tr> <td>3</td><td>8</td><td>Disable 64S-4D trellis encoding.</td></tr> <tr> <td>4</td><td>16</td><td>Disable non-linear coding.</td></tr> </table>	Bit	Value	Result	0	1	Disable 8S-2D trellis encoding.	1	2	Disable 16S-4D trellis encoding.	2	4	Disable 32S-2D trellis encoding.	3	8	Disable 64S-4D trellis encoding.	4	16	Disable non-linear coding.									
Bit	Value	Result																											
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2	4	Disable 32S-2D trellis encoding.																											
3	8	Disable 64S-4D trellis encoding.																											
4	16	Disable non-linear coding.																											

TECHNICAL QUICK REFERENCE

Register	Default	Function		
S34 (cont.)		Bit	Value	Result
		5	32	Disable TX level deviation.
		6	64	Disable Pre-emphasis.
		7	128	Disable Pre-coding.
S35		Reserved		
S36-S37		Reserved		
S38	0	Sets an optional delay, in seconds, before a forced hang-up and clearing of the Transmit buffer when DTR drops during an ARQ call. This allows time for a remote modem to acknowledge receipt of all transmitted data before it is disconnected. The modem immediately hangs up when DTR drops.		
This option only applies to connections terminated by dropping DTR. If the modem receives the ATH command, it ignores S38 and immediately hangs up.				

TECHNICAL QUICK REFERENCE

Register	Default	Function						
S39-S40	Reserved							
S41	0	Bit-mapped register setup. To set registers, see instructions for S13.						
		<table> <tr> <th>Bit</th><th>Value</th><th>Result</th></tr> <tr> <td>0</td><td>1</td><td>Distinctive ring enabled.</td></tr> </table>	Bit	Value	Result	0	1	Distinctive ring enabled.
Bit	Value	Result						
0	1	Distinctive ring enabled.						
S42	0	Reserved						

TECHNICAL QUICK REFERENCE

The Serial Interface

The serial interface is a standard developed by the Electronic Industries Association (EIA). It defines the signals and voltages used when data is exchanged between a computer and a modem or serial printer.

The entire standard covers many more functions than are used in most data communications applications. Data is transmitted between the devices over a shielded serial cable with a 25-pin male (DB-25) connector to the modem and a 25-pin, 9-pin, 8-pin, or custom-built connector to the computer.

FCC regulations require the use of a shielded cable when connecting a modem to a computer to ensure minimal interference with radio and television.

Pin assignments are factory-set in the Sportster modem to match the standard DB-25 assignments in the following table. DB-9 connectors for IBM/AT-compatible computers should be wired at the computer end of the cable as shown in the DB-9 column.

TECHNICAL QUICK REFERENCE

Serial Interface Pin Definitions

DB-25	DB-9	Circuit	Function	Signal Source Computer/Modem
1	—	AA	Chassis Ground	Both
2	3	BA	Transmitted Data	Computer
3	2	BB	Received Data	Modem
4	7	CA	Request to Send	Computer
5	8	CB	Clear to Send	Modem
6	6	CC	Data Set Ready	Modem
7	5	AB	Signal Ground	Both
8	1	CF	Carrier Detect	Modem
12	—	SCF	Speed Indicate	Modem
20	4	CD	Data Terminal Ready	Computer
22	9	CE	Ring Indicate	Modem

REGULATORY INFORMATION

Manufacturer's Declaration of Conformity

U.S. Robotics
7770 North Frontage Road
Skokie, Illinois 60077-2690
U.S.A.

declares that the product *U.S. Robotics Sportster 56K Faxmodem* conforms to the FCC's specifications:

Part 15:

Operation is subject to the following two conditions:

- (1) this device may not cause harmful electromagnetic interference, and
- (2) this device must accept any interference

received including interference that may cause undesired operations.

Part 68:

This equipment complies with FCC Rules Part 68. Located on the bottom of the modem is the FCC Registration Number and Ringer Equivalence Number (REN). You must provide this information to the telephone company when requested.

The REN is used to determine the number of devices you may legally connect to your telephone line. In most areas, the sum of the REN of all devices connected to one line must not exceed five (5.0). You should contact your telephone company to determine the maximum REN for your calling area.

REGULATORY INFORMATION

This equipment uses the following USOC jacks: RJ11C.

This equipment may not be used on coin service provided by the telephone company. Connection to party lines is subject to state tariffs.

An FCC compliant telephone cord and modular plug are provided with this equipment, which is designed to connect to the telephone network or premises wiring using a Part 68 compliant compatible jack. See installation instructions for details.

Caution to the User

The user is cautioned that any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

160

IC (Canada)

This digital apparatus does not exceed the Class B limits for radio noise emissions from digital apparatus set out in the interference-causing equipment standard entitled *Digital Apparatus*, ICES-003 of Industry Canada. Cet appareil numérique respecte les limites de bruits radioélectriques applicables aux appareils numériques de Classe B prescrites dans la norme sur le matériel brouilleur: *Appareils Numériques*, NMB-003 édictée par l'Industrie Canada.

UL Listing/CUL Listing

This information technology equipment is UL-Listed and CUL-Listed for use with UL-Listed personal computers that have installation instructions detailing user installation of card cage accessories.

REGULATORY INFORMATION

Connecting to the Telephone Company

It is not necessary to notify the telephone company before installing the modem. However, the telephone company may request the telephone number(s) to which the Sportster modem is connected and the FCC information printed on this page.

Be sure that the telephone line you are connecting the modem to is a standard analog line and not a digital (PBX), party, or coin telephone line.

If the modem is malfunctioning, it may affect the telephone lines. In this case, disconnect the modem until the source of the difficulty is traced.

Fax Branding

The Telephone Consumer Protection Act of 1991 makes it unlawful for any person to use a computer or other electronic device, including fax machines, to send any message unless such message clearly contains in a margin at the top or bottom of each transmitted page or on the first page of the transmission, the date and time it is sent, an identification of the business or other entity, or other individual sending the message, and the telephone number of the sending machine or of such business, other entity, or individual. (The telephone number provided may not be a 900 number or any other number for which charges exceed local or long-distance transmission charges.)

In order to program this information into your Sportster modem, refer to the RapidComm manual on the CD-ROM that shipped with

REGULATORY INFORMATION

your modem. If you're using a different communications software program, refer to its manual.

Radio and Television Interference

This equipment generates and uses radio frequency energy and if not installed and used properly, in strict accordance with the manufacturer's instructions, may cause interference to radio and television reception. The modem has been tested and found to comply with the limits for a Class B computing device in accordance with the specifications in Part 15 of FCC rules, which are designed to provide reasonable protection against such interference in a residential installation.

However, there is no guarantee that interference will not occur in a particular installation. If this device does cause

interference to radio or television reception, which you can determine by monitoring reception when the modem is installed and when it is removed from the computer, try to correct the problem with one or more of the following measures:

- Re-orient the receiving antenna (for televisions with antenna reception only) or cable input device.
- Relocate the computer with respect to the receiver.
- Relocate the computer and/or the receiver so that they are on separate branch circuits.

If necessary, consult your dealer or an experienced radio/television technician for additional suggestions. You may find the following booklet, prepared by the Federal Communications Commission, helpful:

REGULATORY INFORMATION

How to Identify and Resolve Radio-TV Interference Problems

Stock No. 004-000-0345-4

U.S. Government Printing Office

Washington, DC 20402

In accordance with Part 15 of the FCC rules, the user is cautioned that any changes or modifications to the equipment described in this manual that are not expressly approved by U.S. Robotics, Inc. could void the user's authority to operate the equipment.

For Canadian Modem Users

NOTICE: The Industry Canada (IC) label identifies certified equipment. This certification means the equipment meets certain telecommunications network protective, operational, and safety

requirements as prescribed in the appropriate Terminal Equipment Technical Requirements document(s). The Department does not guarantee the equipment will operate to the user's satisfaction.

Before installing this equipment, users should ensure that it is permissible to be connected to the facilities of the local telecommunications company. The equipment must also be installed using an acceptable method of connection. In some cases, the company's inside wiring associated with a single-line, individual service may be extended by means of a certified connector assembly (telephone extension cord.) The customer should be aware that compliance with the above conditions may not prevent degradation of service in some situations. Currently, telecommunication companies do not allow

REGULATORY INFORMATION

users to connect their equipment to jacks except in precise situations that are spelled out in tariffing arrangements with those companies.

Repairs to certified equipment should be coordinated by a representative designated by the supplier. Any repairs or alterations made by the user to this equipment, or equipment malfunctions, may give the telecommunications company cause to request the user to disconnect the equipment.

For your own protection, make sure that the electrical ground connections of the power utility, telephone lines, and internal metallic water pipe system, if present, are connected together. This precaution may be particularly important in rural areas.

CAUTION: Do **NOT** attempt to make such connections yourself. Instead, contact an

electric inspection authority or electrician, as appropriate.

NOTICE: The Ringer Equivalence Number (REN) assigned to each terminal device provides an indication of the maximum number of terminals allowed to be connected to a telephone interface. The termination on an interface may consist of any combination of devices subject only to the requirement that the sum of the Ringer Equivalence Numbers of all the devices does not exceed 5.

The Ringer Equivalence Number is located on the bottom of the modem's case (external modems) or on the modem's circuit board (internal modems).

WARRANTY AND REPAIR SERVICE
CENTER:

REGULATORY INFORMATION

Keating Technologies
25 Royal Crest Court, Suite 200
Markham, ONT L3R 9X4

AVIS: L'étiquette de Industrie Canada identifie le matériel homologué. Cette étiquette certifie que le matériel est conforme à certaines normes de protection, d'exploitation et de sécurité des réseaux de télécommunications. Le Ministère n'assure toutefois pas que le matériel fonctionnera à la satisfaction de l'utilisateur.

Avant d'installer ce matériel, l'utilisateur doit s'assurer qu'il est permis de le raccorder aux installations de l'entreprise locale de télécommunication. Le matériel doit également être installé en suivant une méthode acceptée de raccordement. L'abonné ne doit pas oublier qu'il est possible que la conformité aux conditions énoncées ci-dessus n'empêche pas

le dégradation du service dans certaines situations.

Les réparations de matériel homologué doivent être effectuées par un centre d'entretien canadien autorisé désigné par le fournisseur. La compagnie de télécommunications peut demander à l'utilisateur de débrancher un appareil à la suite de réparations ou de modifications effectuées par l'utilisateur ou à cause de mauvais fonctionnement.

Pour sa propre protection, l'utilisateur doit s'assurer que tous les fils de mise à la terre de la source d'énergie électrique, des lignes téléphoniques et des canalisations d'eau métalliques, s'il y en a, sont raccordé ensemble. Cette précaution est particulièrement importante dans les régions rurales.

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Avertissement: L'utilisateur ne doit pas tenter de faire ces raccordements lui-même; il doit avoir recours à un service d'inspection des installations électriques, ou à un électricien, selon le cas.

NOTICE: L'Indice d'Equivalence de la Sonnerie (IES) de chaque appareil donne une indication du nombre maximal de terminaux qui peut être branché à l'interface téléphonique. La termination d'une interface peut consister de n'importe quelle combinaison d'appareils sur le réseau, seulement si la somme des IES de tous les appareils n'excède pas 5.

L'Indice d'Equivalence de la Sonnerie (IES) est situé au revers du modem (modems externes), ou sur le circuit imprimé (modems internes).

Centre de garantie et de service après-vente:

Keating Technologies
25 Royal Crest Court, Suite 200
Markham, ONT L3R 9X4

Limited Warranty

U.S. Robotics Access Corp. warrants to the original end-user purchaser that this product will be free from defects in materials and workmanship for a period of five years from the date of purchase. During the limited warranty period, and upon proof of purchase, the product will be repaired or replaced (with the same or a similar model, which may be a refurbished model) at U.S. Robotics' option, without charge for either parts or labor. This limited warranty shall not apply if the product is modified, tampered with, misused, or subjected to

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abnormal working conditions (including, but not limited to, lightning and water damage).

THIS LIMITED WARRANTY DOES NOT GUARANTEE YOU UNINTERRUPTED SERVICE. REPAIR OR REPLACEMENT AS PROVIDED UNDER THIS LIMITED WARRANTY IS THE EXCLUSIVE REMEDY OF THE PURCHASER. THIS LIMITED WARRANTY IS IN LIEU OF ALL OTHER WARRANTIES, EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, ANY IMPLIED WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR USE OR PURPOSE. U.S. ROBOTICS SHALL IN NO EVENT BE LIABLE FOR ANY SPECIAL, INDIRECT, INCIDENTAL,

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above exclusion or limitation may not apply to you.

To obtain service under this limited warranty, contact the U.S. Robotics Technical Support Department at 847-982-5151 or by mail at 3Com, 7770 N. Frontage Road, Attn.: Technical Support Dept., Skokie, Illinois 60077-2690. You will be given a Service Repair Order (“SRO”) number to help U.S. Robotics keep track of your limited warranty request. Once you have received your SRO number, take or send the product, postage prepaid and insured, to 3Com, Attn: RMA, [your SRO#], 6201 W. Oakton, East Dock, Morton Grove, IL 60053. Pack the modem in a strong corrugated cardboard box with plenty of packing material. DO NOT send the modem back in its original box. DO NOT send anything but the modem (do not send

back the power supply, CD-ROM, documentation, etc.). If possible, send the modem via a courier capable of tracking the progress of the shipment. Include proof of the date of purchase. IMPORTANT: If you send your unit, pack it securely, and be sure that your SRO number is visible on the outside of the package.

Index

A

America Online, U.S. Robotics online support and, 108-109
American Standard Code for Information Interchange. *See* ASCII
analog loopback, 113
analog phone lines, vs. digital, 18, 26, 33, 37
analog signals, 113
answer mode, 113
AOL. *See* America Online, U.S. Robotics online support and
ARQ. *See* Automatic Repeat reQuest
ASCII (American Standard Code for Information Interchange), 113
asynchronous transmission, 113
AT commands, 125-139
auto answer, 114
auto answer (AA) light, 124

auto dial, 114
Automatic Repeat reQuest (ARQ) light, 124

B

baud rate, 114
BBS, U.S. Robotics, 105-106
binary digit, 114
bit rate, 114
bits per second (bps), 114
buffer, 114
byte, 114

Index

C

- carrier, 115
- carrier detect (CD) light, 124
- character, 115
- characters per second (cps), 115
 - low, errors and, 103
- class 1 and 2.0 faxing standards, 115
- clear to send (CS) light, 124
- COM ports
 - conflicts with IRQs, 99-100
 - and Plug and play installations, 6, 21-22
- COM drivers, outdated, 104
- command summary, 125-139
- communications software
 - and flow control, 61-62
 - installing other fax/data software, 43-44
- 170
 - installing RapidComm, 43-58
 - UART, 62
- CompuServe, U.S. Robotics online support and, 108
- Connections* CD-ROM
 - Modem Station, 63-86
 - Modem Update Wizard, 87-98
 - software installation, 43-58
- CRC. **See** Cyclic Redundancy Checking
- Cyclic Redundancy Checking (CRC), 103, 115

Index

D

data compression table, 115
data mode, 116
data terminal (terminating) equipment (DTE), 116
data terminal ready (TR) light, 124
default settings, 116
detect phase, 116
dictionary, 116
digital loopback, 116
digital signals, 116
DIP switches, 140-141
DTE. **See** data terminal (terminating)
equipment
duplex, 116

E

EIA. **See** Electronic Industries Association

Electronic Industries Association (EIA), 117
error control, 117
 disabling, 102
error control/ (ARQ/) light, 124
external modems
 and faxing, 58-60
 hardware installation
 Windows 3.x, 31-33
 Windows 95, 35-41
 and serial cable, 31-32, 35-36

F

facsimile (fax), 117
fax mode, 117
fax operations (FAX) light, 124
faxing with RapidComm, 58-60
flow control, 61-62, 102, 117
frame, 117

Index

front panel lights, summary, 124
FTP, U.S. Robotics online support and, 108
full duplex, 117

G

glossary, 113-122

H

half duplex, 117
hertz (Hz), 118

I

internal modems
 faxing with, 58-60
 hardware installation
 Windows 3.x, 5-20
 Windows 95, 21-29

172

and Plug and Play installation, 6, 21-22

International Telegraph and Telephone Consultative Committee (ITU-T), 118

IRQs, 5-14

 conflicts, 99-100

ISA expansion slots, 16, 24

ITU-T. **See** International Telegraph and Telephone Consultative Committee

J

jumper pins, 6, 13-14, 22

L

LAPM. **See** Link Access Procedure for Modems

Index

Link Access Procedure for Modems (LAPM),
118

local echo, 118
 turning off, 101

M

Microcom Networking Protocol (MNP), 118

MNP. *See* Microcom Networking Protocol

Modem Station, 63-86

 installing, 63-65

 using, 66-86

Modem Update Wizard, 87-98

 installing, 87-91

 using, 92-98

N

nonvolatile memory (NVRAM), 118

NVRAM. *See* nonvolatile memory

O

off/on hook, 118

online fallback/fall forward, 119

online help resources, 105-109

originate mode, 119

P

parity, 102, 119

Plug and Play installation, 6, 21-22

protocol, 119

R

RAM (random access memory), 119

random access memory. *See* RAM

RapidComm

 faxing with, 58-60

Index

installing, 43-58
troubleshooting, 110-111

read only memory (ROM), 120
received data (RD) light, 124
remote digital loopback, 119
remote echo, 119

S

s-registers, 142-152
send data (SD) light, 124
serial cable, purchasing, 31-32, 35-36
serial interface, 153-154
 pin definitions, 154
serial number, modem, 15, 23
serial (COM) ports, 5-14, 21-22, 32, 36
 conflicts, 99-100
serial transmission, 118
shunts, 13-14, 22

start/stop bits, 102, 105, 120

T

Technical Reference Guide, downloading,
106-108
terminal mode, 105, 120
Terminate and Stay Resident programs, errors
and, 103-104
throughput, 120
troubleshooting, 99-104
 RapidComm, 110-111
TSR. *See* Terminate and Stay Resident
programs

U

UART, 62
upgrading/updating modems, 87-98

Index

U.S. Robotics

BBS, 105-106

online help resources, 105-109

Technical Reference Guide,

downloading, 106-108

Technical Support Hotline, 109

V

V.17 fax transmissions, errors and, 104

W

World Wide Web (WWW), U.S. Robotics

online support and, 108

X

x2 technology, 2-3

Xmodem, 122

xon/xoff, 122

Y

Ymodem, 122

Ymodem G, 122

Z

Zmodem, 122

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